CDN을 활용한 OTT 서비스 구축

TEAM

MEGAFLIX

Leader	조현석
	김세희
	임동규

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- 개요
- 기술 스택

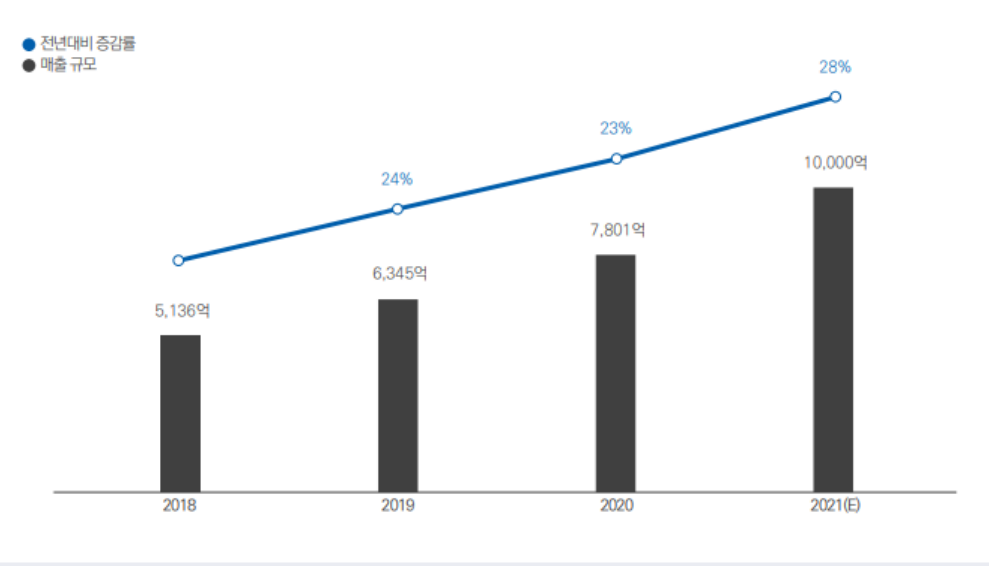
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- 타임라인
- 시스템 아키텍처

3 과정
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- On-Premise
- CI / CD

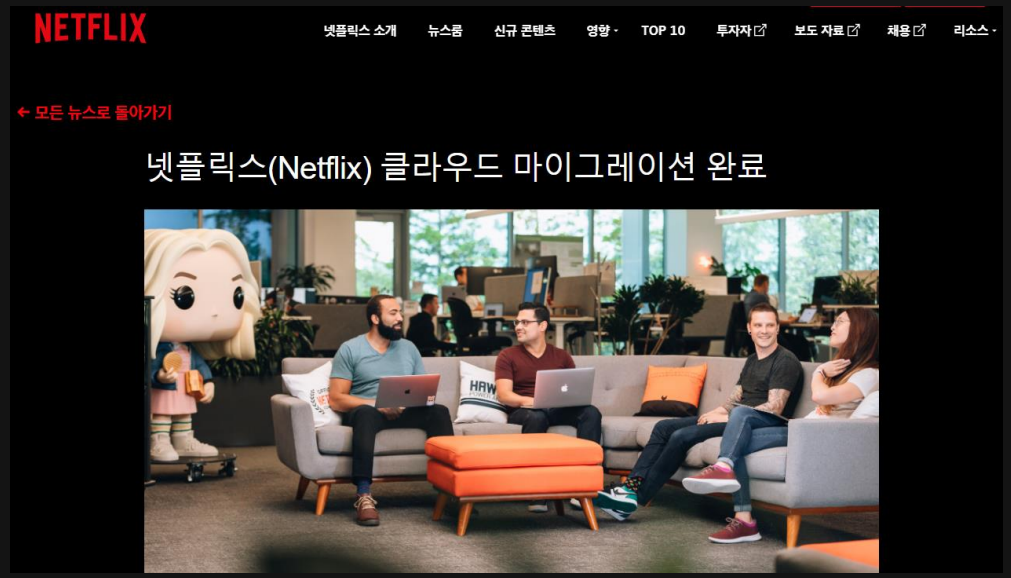
4 결과
- 서비스 사용

개요

국내 OTT 시장 규모



넷플릭스
클라우드 마이그레이션



개요

Client

글로벌 고객들을 대상으로 빠른 속도를 보장하는 스트리밍 서비스를 구축하려고 합니다.

컨텐츠 유출을 최대한 방지할 수 있는 대안이 있으면 좋겠어요. 시스템적으로 문제가 발생해도 고객들은 서비스를 이용할 수 있어야 하고요.

추가로 운영 및 개발 환경의 편의성을 고려했으면 좋겠습니다.

CDN 활용

보안 강화

고가용성

CI/CD 구축

Megaflix



사용 기술 스택

플랫폼

aws

vmware ESXi

온프레미스

GitLab

docker

HARBOR

Jenkins

libreswan

기타 도구

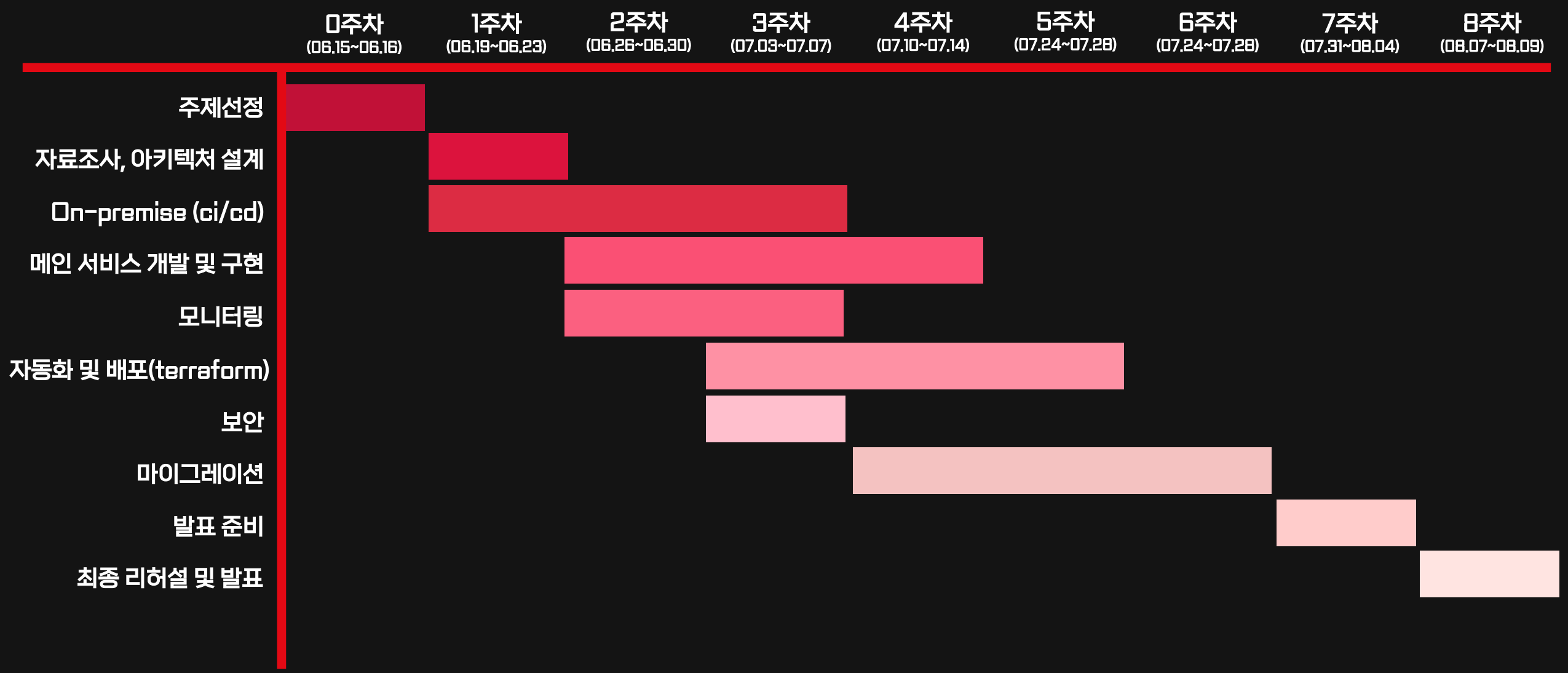
HashiCorp Terraform

kubernetes

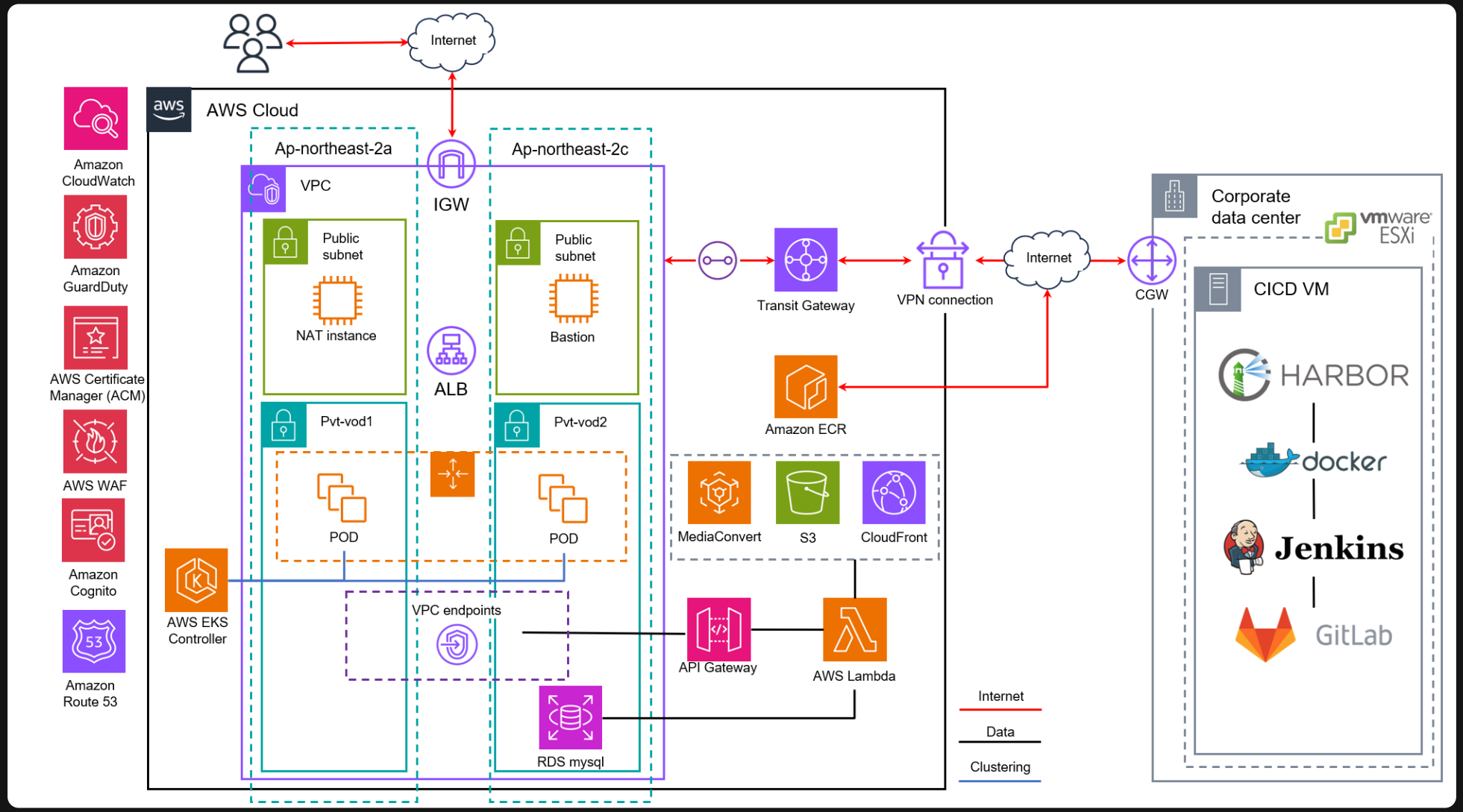
React

NGINX

TIME LINE


0주차
(06.15~06.16)
1주차
(06.19~06.23)
2주차
(06.26~06.30)
3주차
(07.03~07.07)
4주차
(07.10~07.14)
5주차
(07.24~07.28)
6주차
(07.24~07.28)
7주차
(07.31~08.04)
8주차
(08.07~08.09)
주제선정
자료조사, 아키텍처 설계
On-premise (ci/cd)
메인 서비스 개발 및 구현
모니터링
자동화 및 배포(terraform)
보안
마이그레이션
발표 준비
최종 리허설 및 발표

ARCHITECTURE

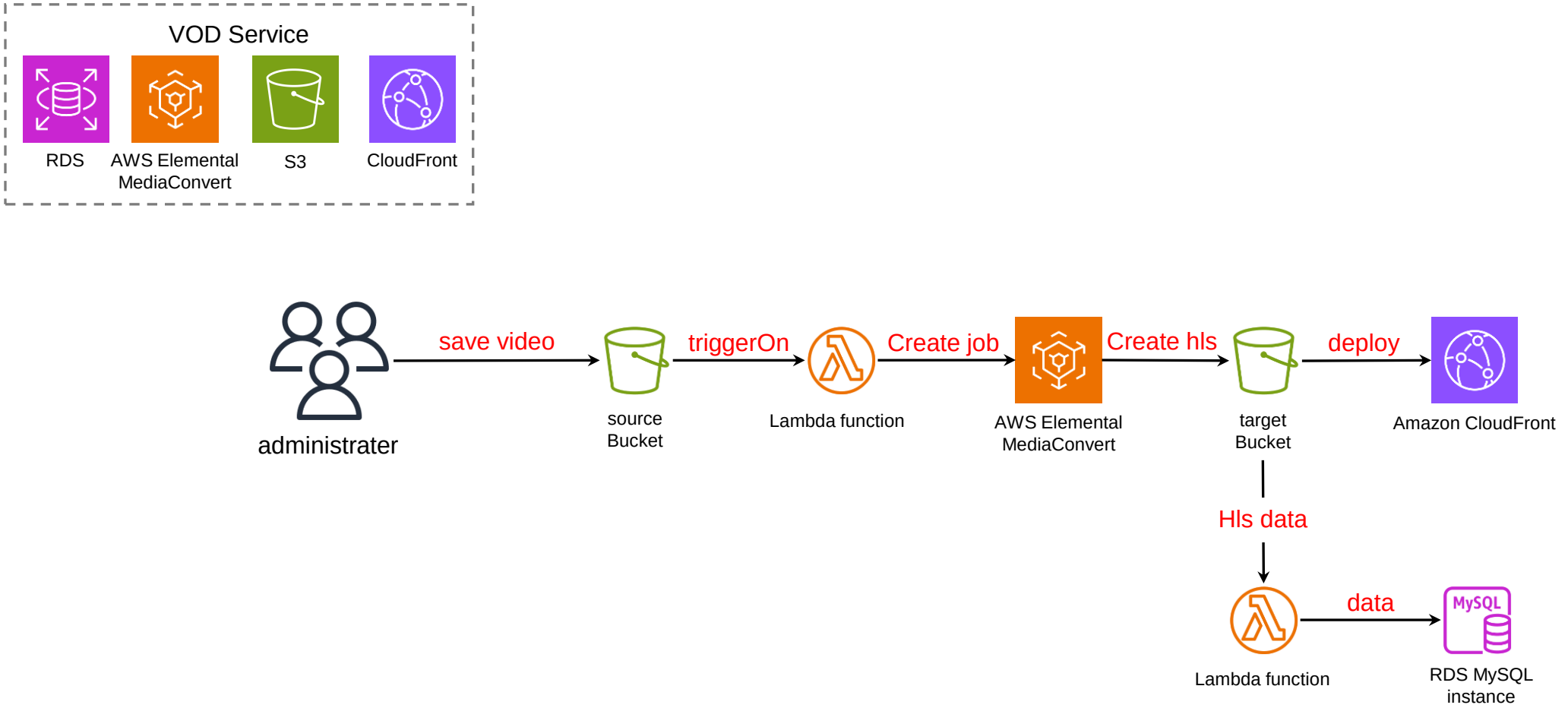




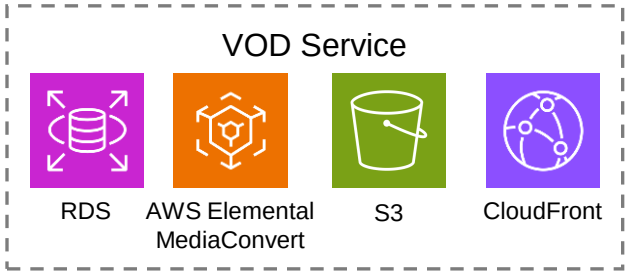
CDN & WEB SERVICE

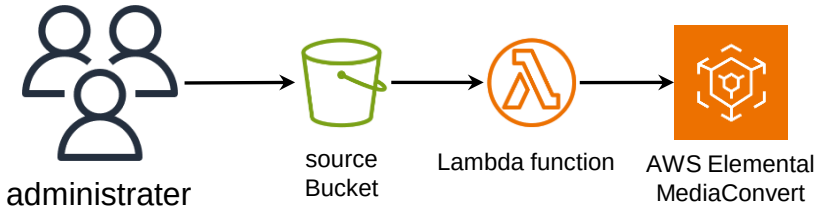
- CLOUDFRONT
- FE & BE

CDN

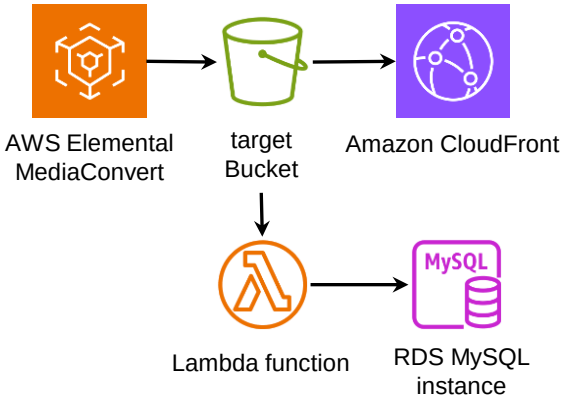

VOD Service
RDS
AWS Elemental MediaConvert
S3
CloudFront
administrater
save video
source Bucket
triggerOn
Lambda function
Create job
AWS Elemental MediaConvert
Create hls
target Bucket
deploy
Amazon CloudFront
Hls data
Lambda function
data
MySQL
RDS MySQL instance

CDN





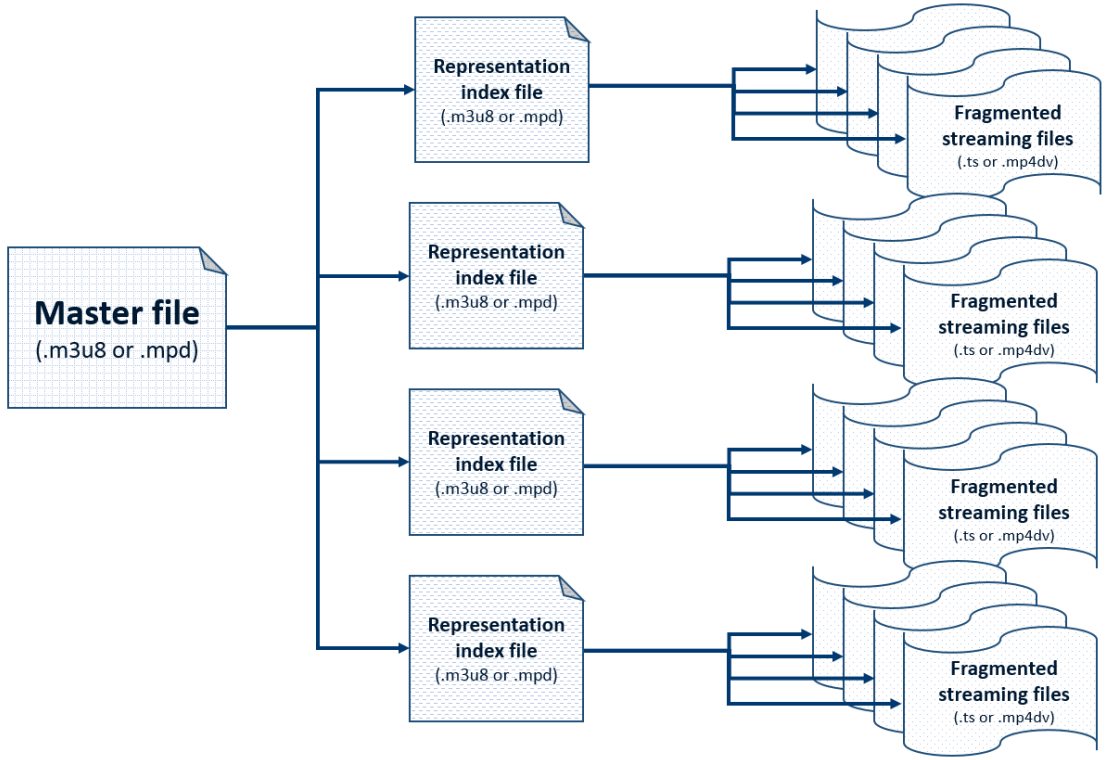
컨텐츠 변환 자동화



DB 데이터 갱신 자동화

VOD Service

AWS Elemental
MediaConvert



HLS란?

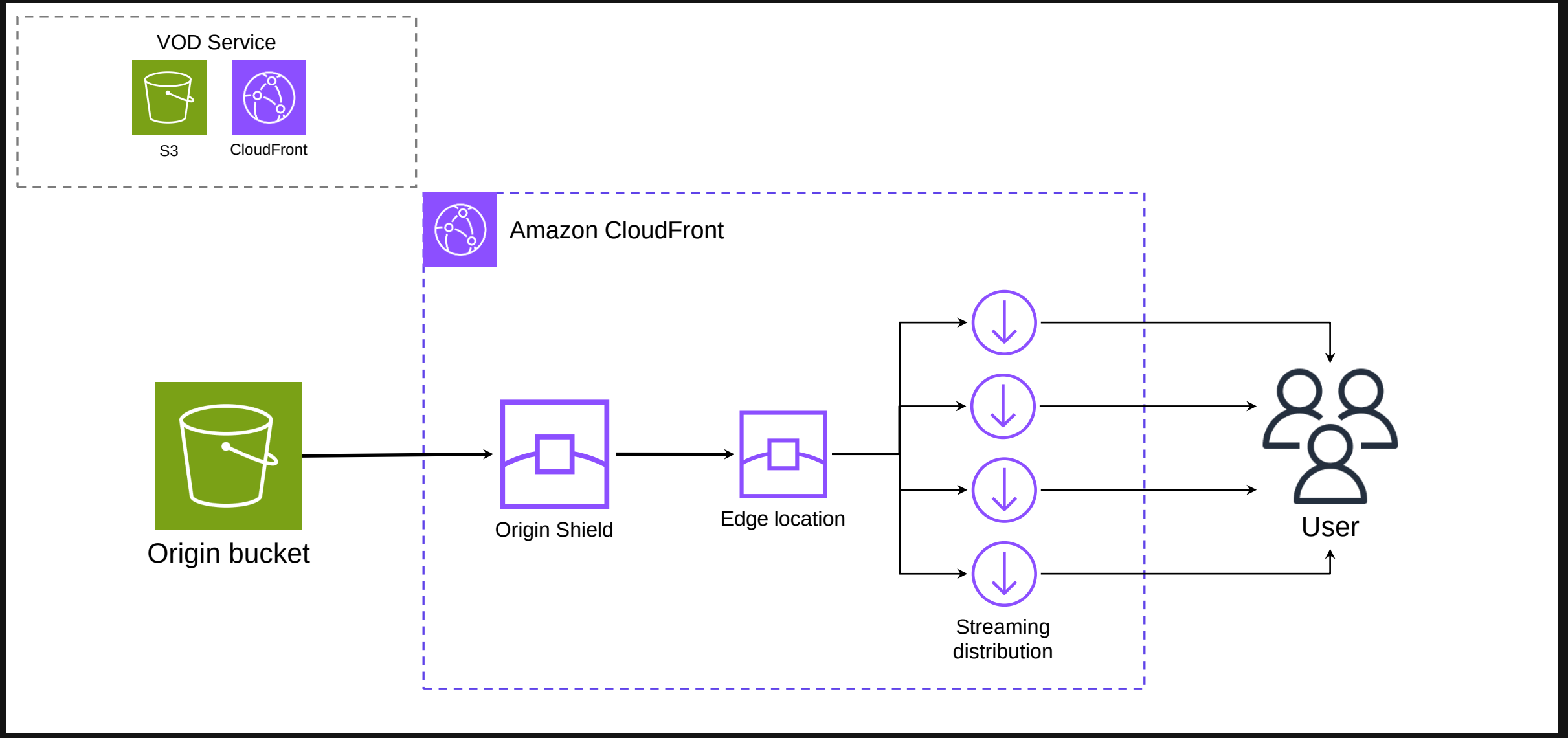
- HTTP으로 영상 스트리밍을 할 때 사용하는 프로토콜
- 네트워크 상태에 따라 영상 화질을 선택가능

HLS Protocol

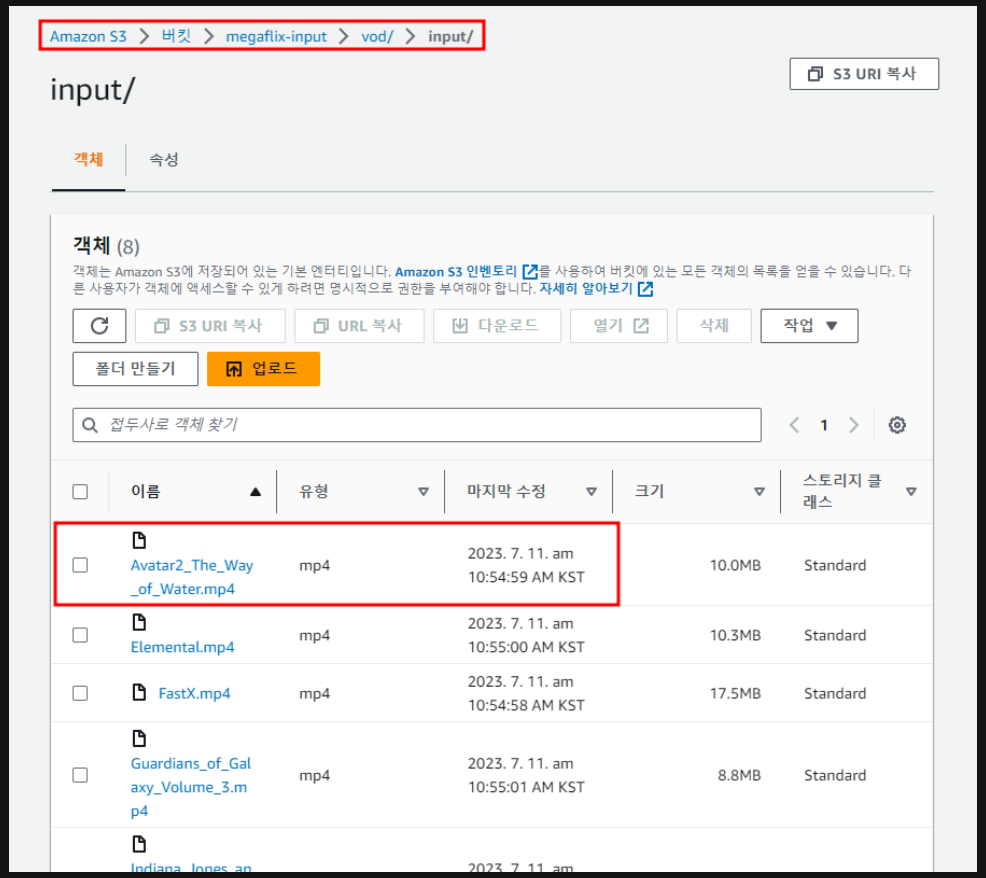
.m3u8 — 영상 재생을 위한 메타 정보

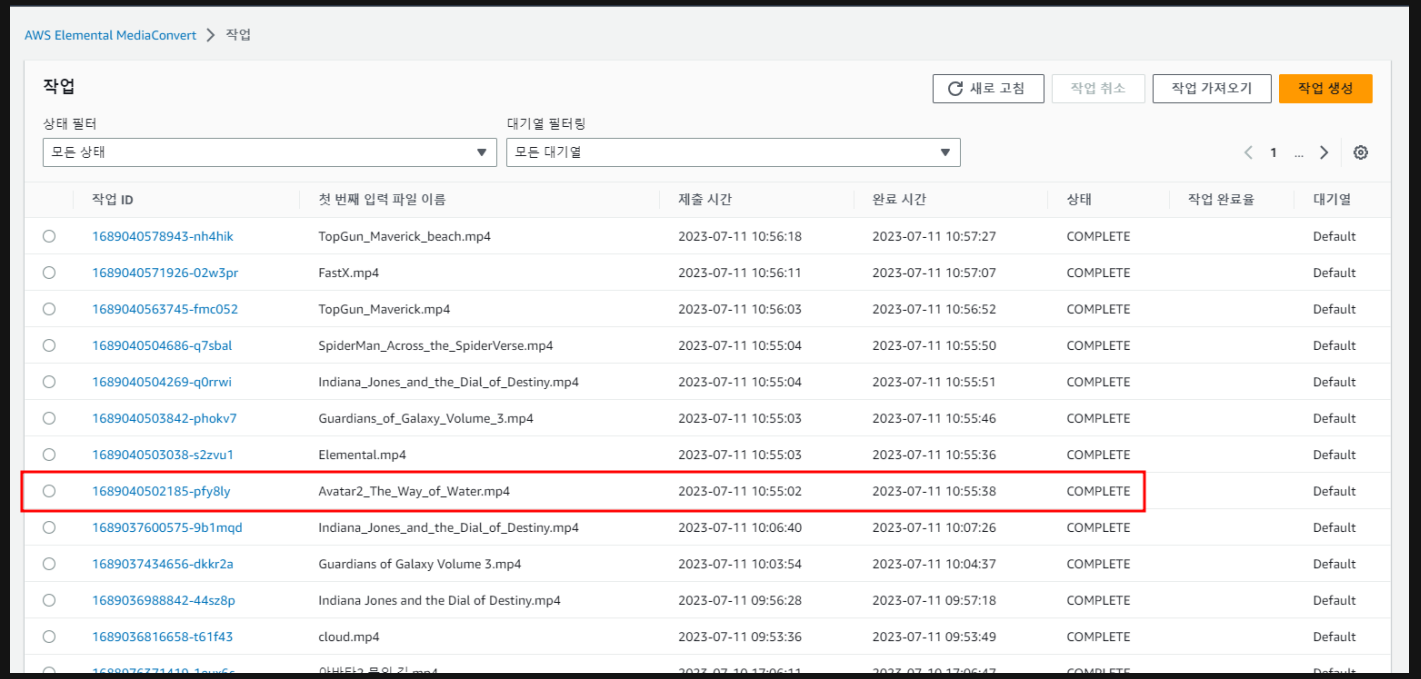
.ts — 시간단위로 나뉘어진 실제 스트리밍 영상 데이터

CDN

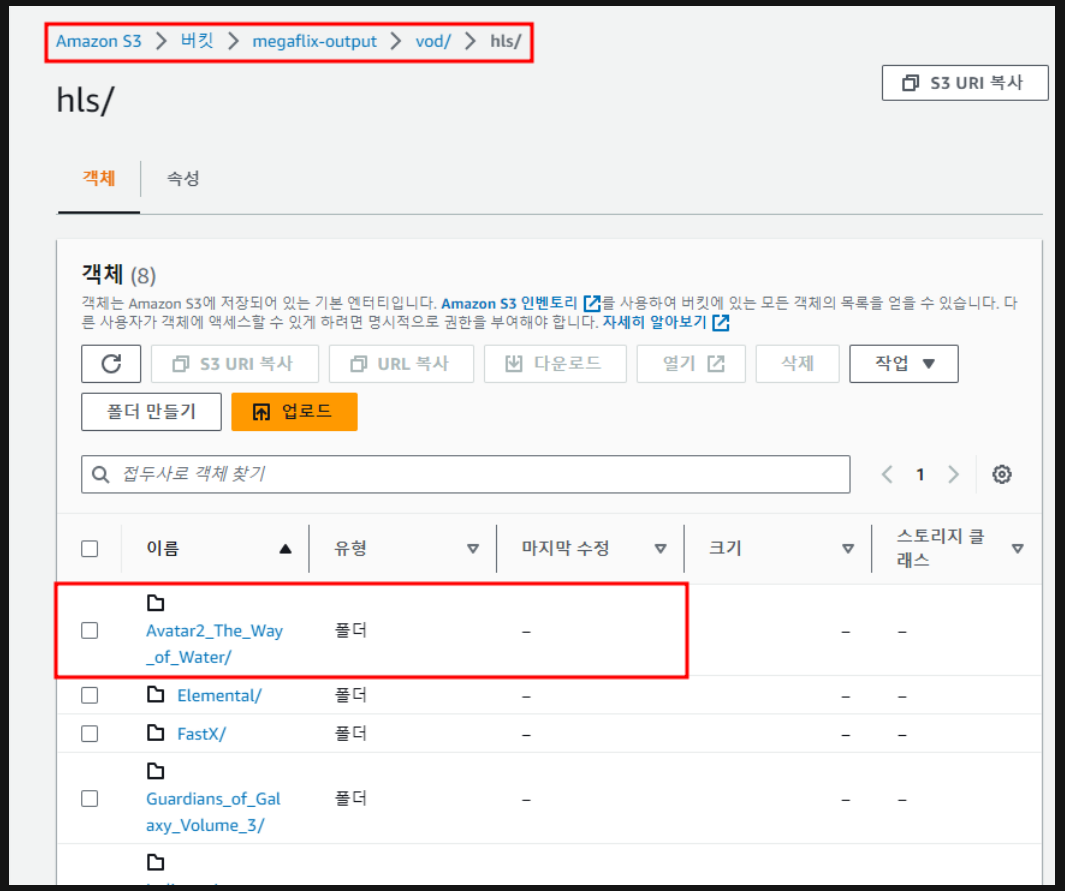

VOD Service
S3
CloudFront
Amazon CloudFront
Origin bucket
Origin Shield
Edge location
Streaming distribution
User

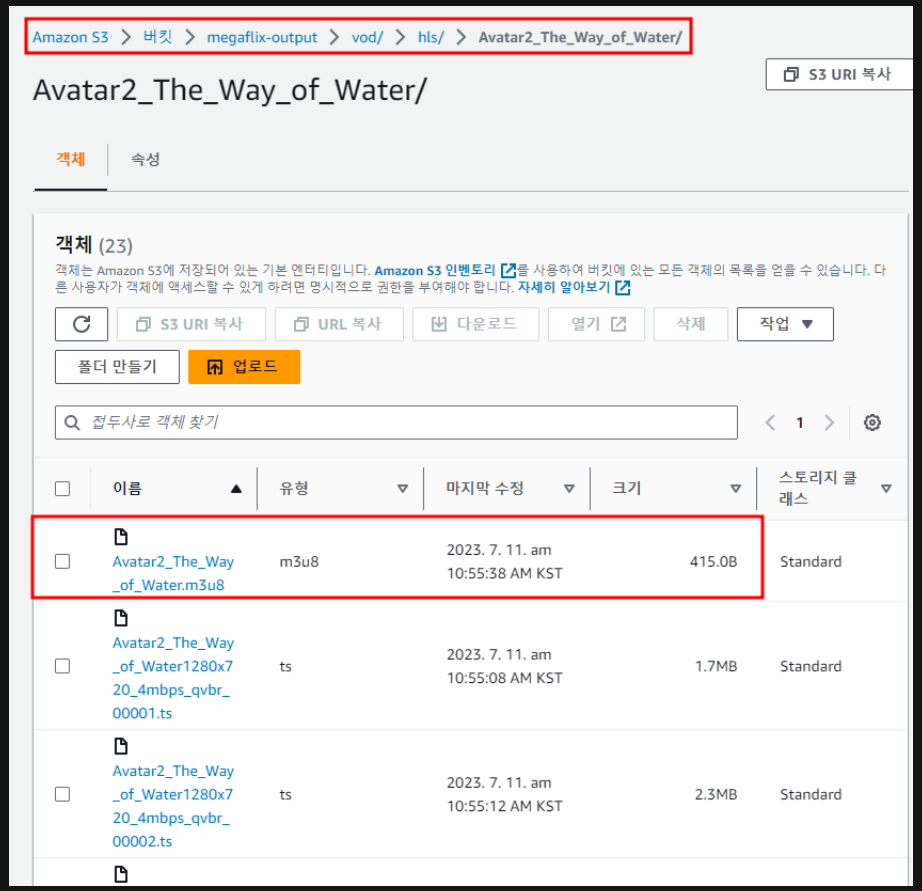
CDN

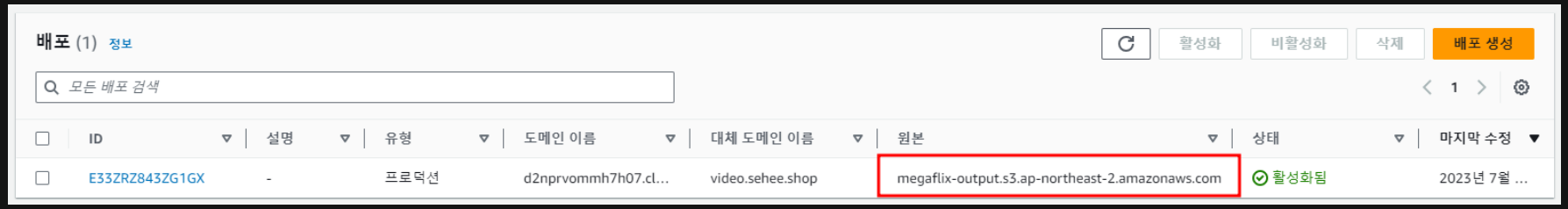

Amazon S3 > 버킷 > megaflix-input > vod/ > input/
input/
S3 URI 복사
객체
속성
객체 (8)
객체는 Amazon S3에 저장되어 있는 기본 엔터티입니다. Amazon S3 인벤토리를 사용하여 버킷에 있는 모든 객체의 목록을 얻을 수 있습니다. 다른 사용자가 객체에 액세스할 수 있게 하려면 명시적으로 권한을 부여해야 합니다. 자세히 알아보기
S3 URI 복사
URL 복사
다운로드
열기
삭제
작업
폴더 만들기
업로드
접두사로 객체 찾기
이름
유형
마지막 수정
크기
스토리지 클래스
Avatar2_The_Way_of_Water.mp4 mp4 2023. 7. 11. am 10:54:59 AM KST 10.0MB Standard
Elemental.mp4 mp4 2023. 7. 11. am 10:55:00 AM KST 10.3MB Standard
FastX.mp4 mp4 2023. 7. 11. am 10:54:58 AM KST 17.5MB Standard
Guardians_of_Galaxy_Volume_3.mp4 mp4 2023. 7. 11. am 10:55:01 AM KST 8.8MB Standard


AWS Elemental MediaConvert > 작업
작업
새로 고침
작업 취소
작업 가져오기
작업 생성
상태 필터
모든 상태
대기열 필터링
모든 대기열
작업 ID
첫 번째 입력 파일 이름
제출 시간
완료 시간
상태
작업 완료율
대기열
1689040578943-nh4hik TopGun_Maverick_beach.mp4 2023-07-11 10:56:18 2023-07-11 10:57:27 COMPLETE Default
1689040571926-02w3pr FastX.mp4 2023-07-11 10:56:11 2023-07-11 10:57:07 COMPLETE Default
1689040563745-fmc052 TopGun_Maverick.mp4 2023-07-11 10:56:03 2023-07-11 10:56:52 COMPLETE Default
1689040504686-q7sbal SpiderMan_Across_the_SpiderVerse.mp4 2023-07-11 10:55:04 2023-07-11 10:55:50 COMPLETE Default
1689040504269-q0rrwi Indiana_Jones_and_the_Dial_of_Destiny.mp4 2023-07-11 10:55:04 2023-07-11 10:55:51 COMPLETE Default
1689040503842-phokv7 Guardians_of_Galaxy_Volume_3.mp4 2023-07-11 10:55:03 2023-07-11 10:55:46 COMPLETE Default
1689040503038-s2zvu1 Elemental.mp4 2023-07-11 10:55:03 2023-07-11 10:55:36 COMPLETE Default
1689040502185-pfy8ly Avatar2_The_Way_of_Water.mp4 2023-07-11 10:55:02 2023-07-11 10:55:38 COMPLETE Default
1689037600575-9b1mqd Indiana_Jones_and_the_Dial_of_Destiny.mp4 2023-07-11 10:06:40 2023-07-11 10:07:26 COMPLETE Default
1689037434656-dkkr2a Guardians of Galaxy Volume 3.mp4 2023-07-11 10:03:54 2023-07-11 10:04:37 COMPLETE Default
1689036988842-44sz8p Indiana Jones and the Dial of Destiny.mp4 2023-07-11 09:56:28 2023-07-11 09:57:18 COMPLETE Default
1689036816658-t61f43 cloud.mp4 2023-07-11 09:53:36 2023-07-11 09:53:49 COMPLETE Default

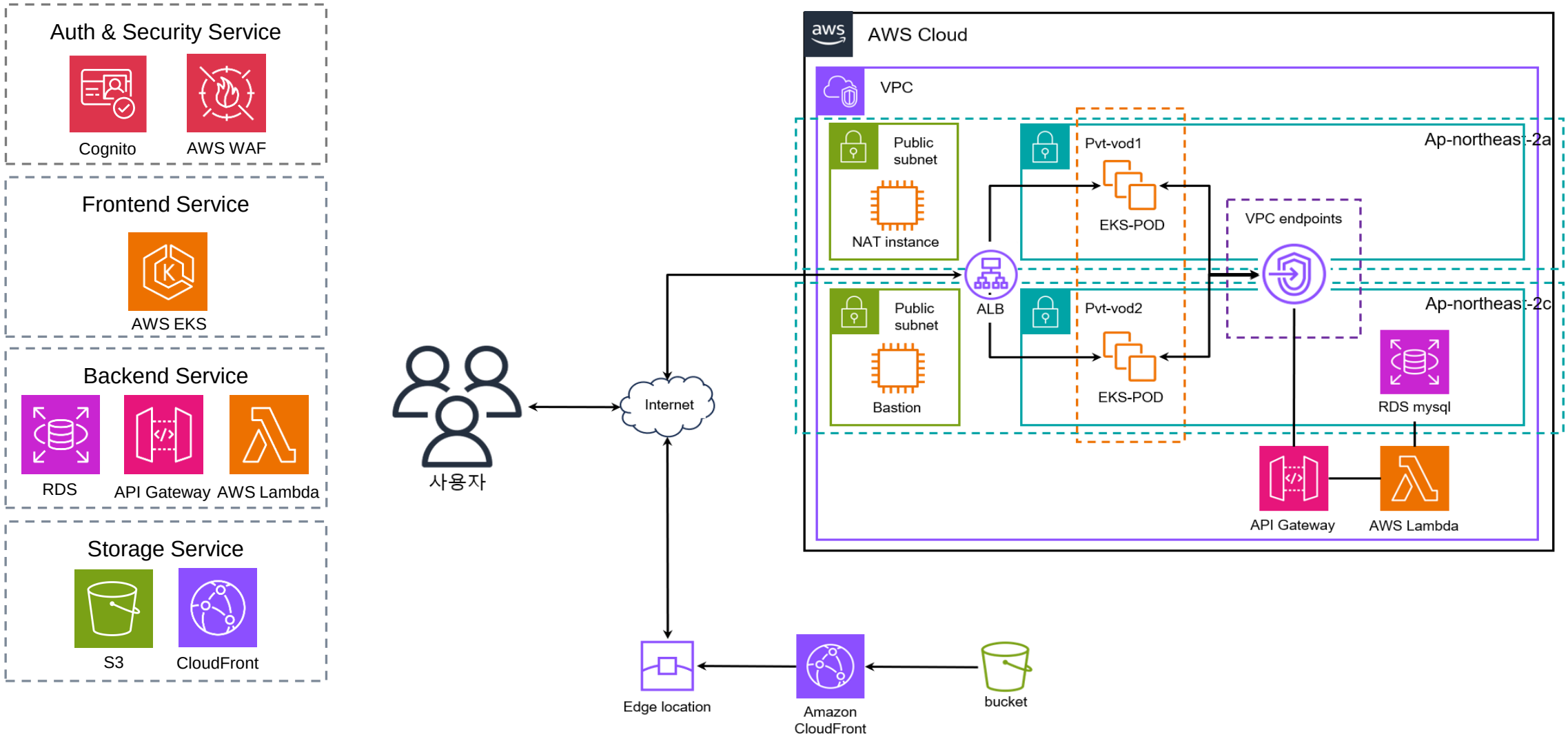
CDN


Amazon S3 > 버킷 > megaflix-output > vod/ > hls/
hls/
S3 URI 복사
객체
속성
객체 (8)
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S3 URI 복사
URL 복사
다운로드
열기
삭제
작업
폴더 만들기
업로드
접두사로 객체 찾기
1
이름
유형
마지막 수정
크기
스토리지 클래스
Avatar2_The_Way_of_Water/
폴더
Elemental/
폴더
FastX/
폴더
Guardians_of_Galaxy_Volume_3/
폴더

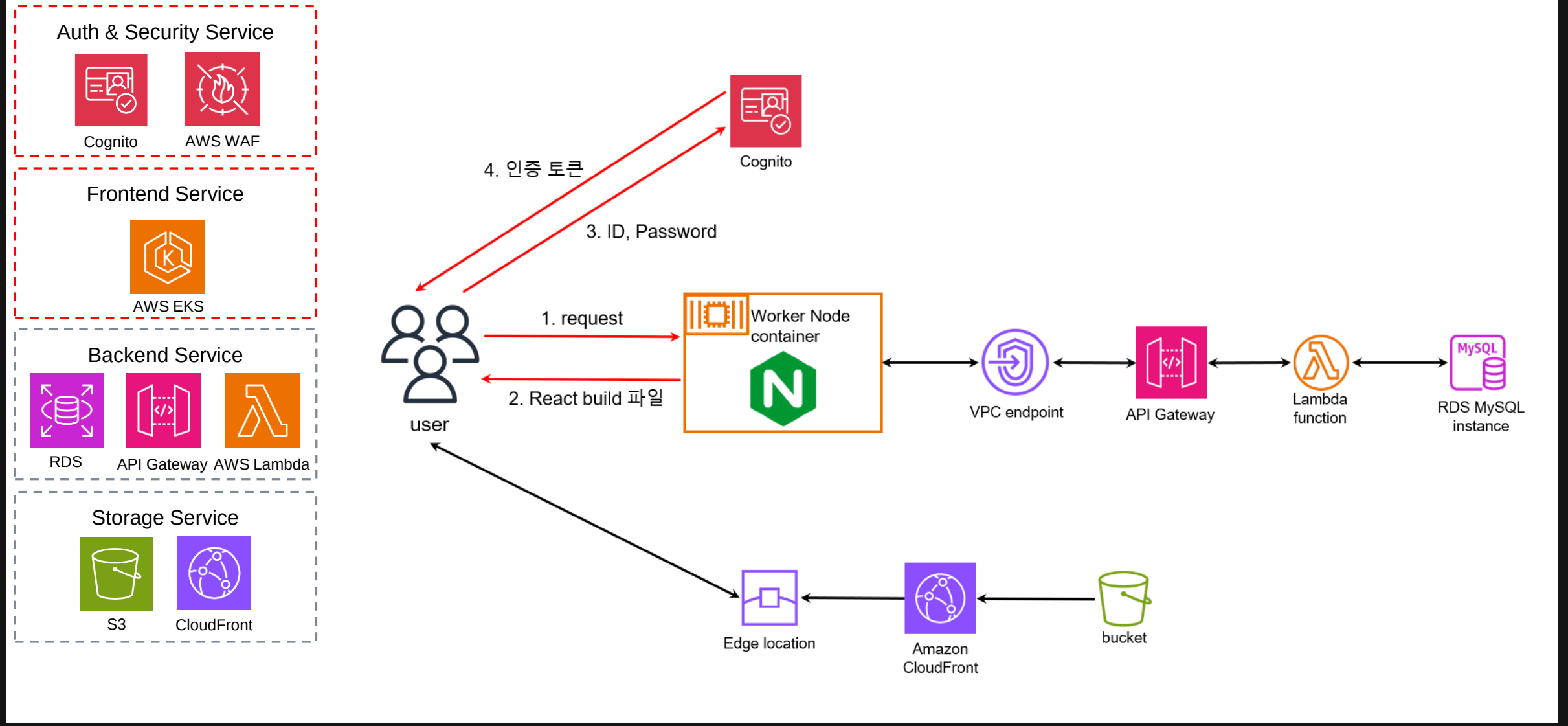

Amazon S3 > 버킷 > megaflix-output > vod/ > hls/ > Avatar2_The_Way_of_Water/
Avatar2_The_Way_of_Water/
S3 URI 복사
객체
속성
객체 (23)
객체는 Amazon S3에 저장되어 있는 기본 엔터티입니다. Amazon S3 인벤토리를 사용하여 버킷에 있는 모든 객체의 목록을 얻을 수 있습니다. 다른 사용자가 객체에 액세스할 수 있게 하려면 명시적으로 권한을 부여해야 합니다. 자세히 알아보기
S3 URI 복사
URL 복사
다운로드
열기
삭제
작업
폴더 만들기
업로드
접두사로 객체 찾기
1
이름
유형
마지막 수정
크기
스토리지 클래스
Avatar2_The_Way_of_Water.m3u8
m3u8
2023. 7. 11. am 10:55:38 AM KST
415.0B
Standard
Avatar2_The_Way_of_Water1280x720_4mbps_qvbr_00001.ts
ts
2023. 7. 11. am 10:55:08 AM KST
1.7MB
Standard
Avatar2_The_Way_of_Water1280x720_4mbps_qvbr_00002.ts
ts
2023. 7. 11. am 10:55:12 AM KST
2.3MB
Standard


배포 (1) 정보
활성화
비활성화
삭제
배포 생성
모든 배포 검색
1
ID
설명
유형
도메인 이름
대체 도메인 이름
원본
상태
마지막 수정
E33ZRZ843ZG1GX
-
프로덕션
d2nprvommh7h07.cl...
video.sehee.shop
megaflix-output.s3.ap-northeast-2.amazonaws.com
활성화됨
2023년 7월 ...

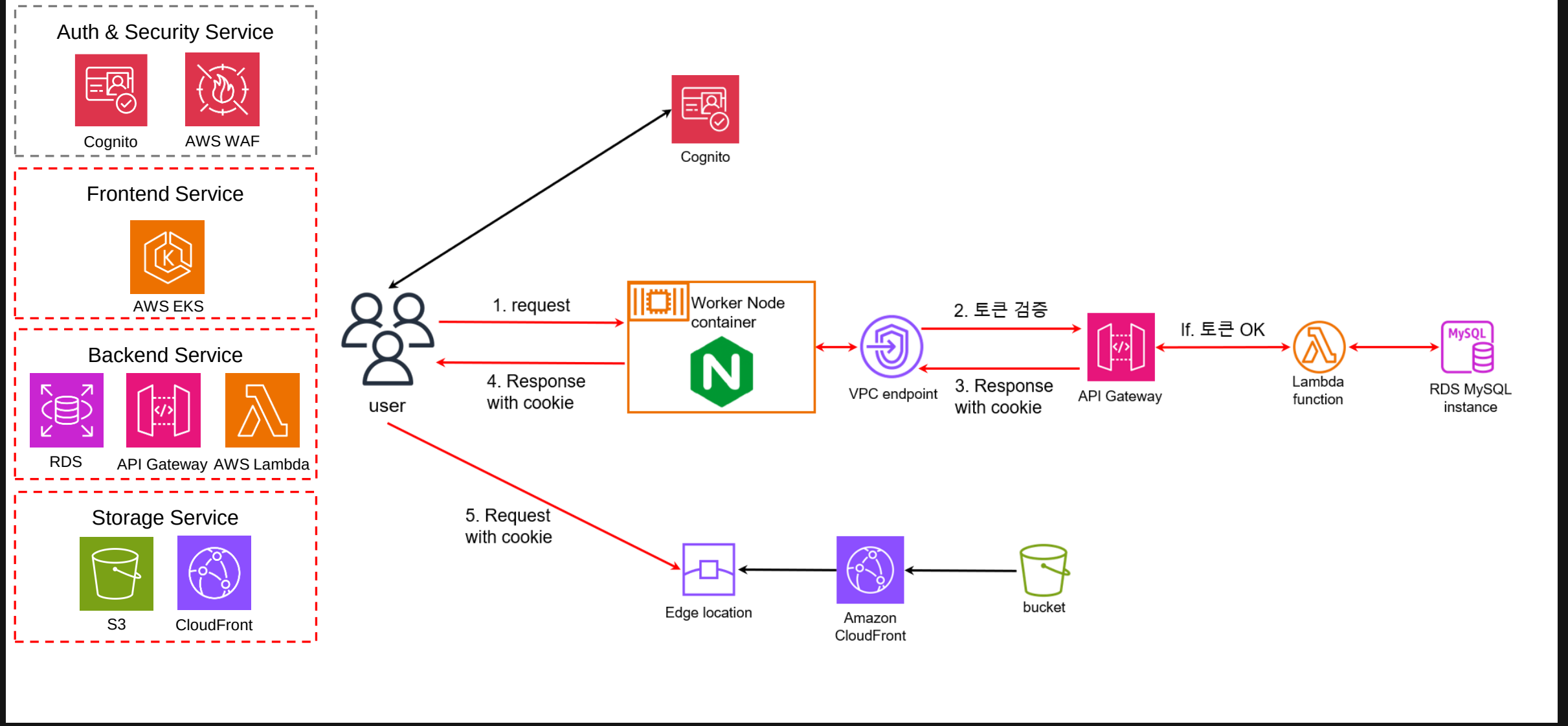
WEB SERVICE


Auth & Security Service
Cognito
AWS WAF
Frontend Service
AWS EKS
Backend Service
RDS
API Gateway
AWS Lambda
Storage Service
S3
CloudFront
사용자
Internet
aws
AWS Cloud
VPC
Public subnet
NAT instance
Public subnet
Bastion
ALB
Pvt-vod1
EKS-POD
Pvt-vod2
EKS-POD
VPC endpoints
Ap-northeast-2a
Ap-northeast-2c
RDS mysql
API Gateway
AWS Lambda
Edge location
Amazon CloudFront
bucket

WEB SERVICE

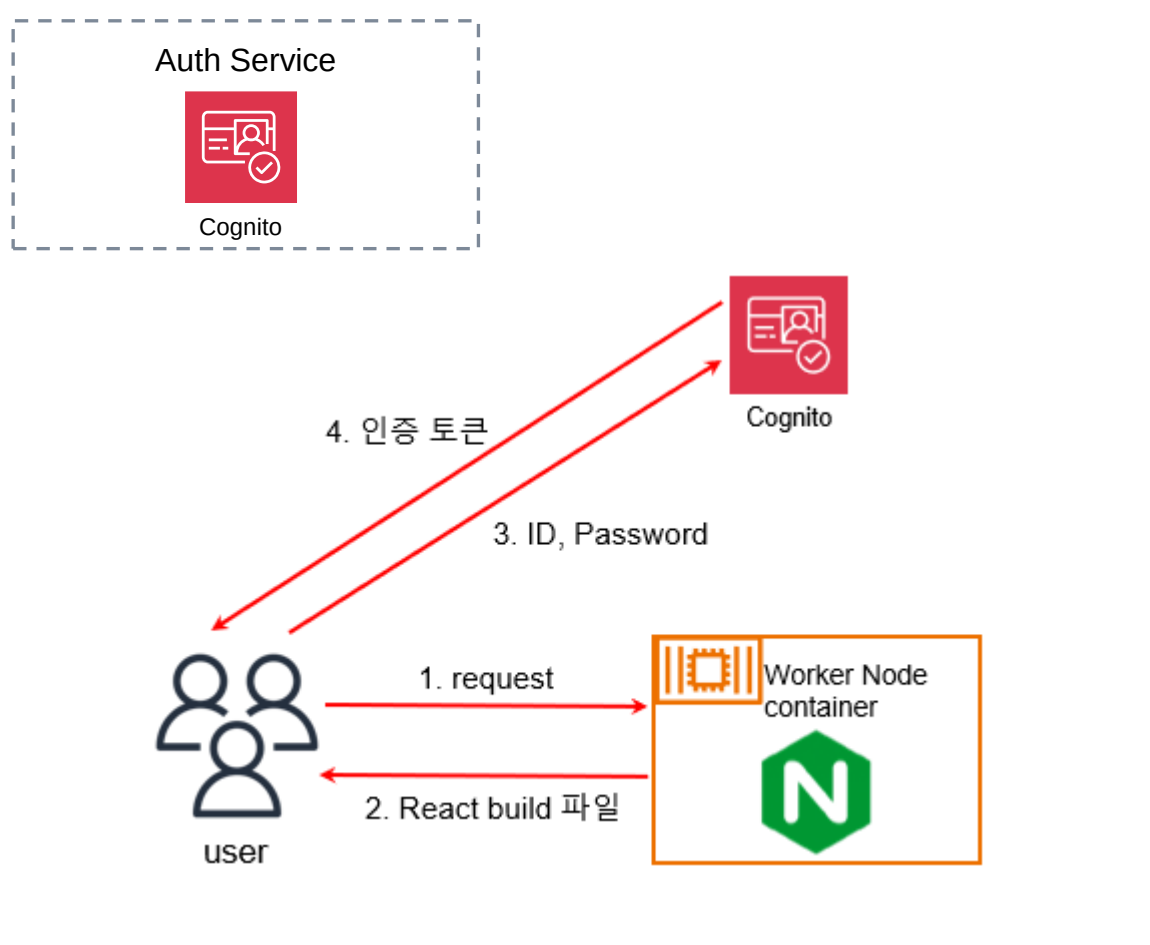

Auth & Security Service
Cognito
AWS WAF
Frontend Service
AWS EKS
Backend Service
RDS
API Gateway
AWS Lambda
Storage Service
S3
CloudFront
Cognito
4. 인증 토큰
3. ID, Password
user
1. request
2. React build 파일
Worker Node
container
VPC endpoint
API Gateway
Lambda
function
MySQL
RDS MySQL
instance
Edge location
Amazon
CloudFront
bucket

WEB SERVICE

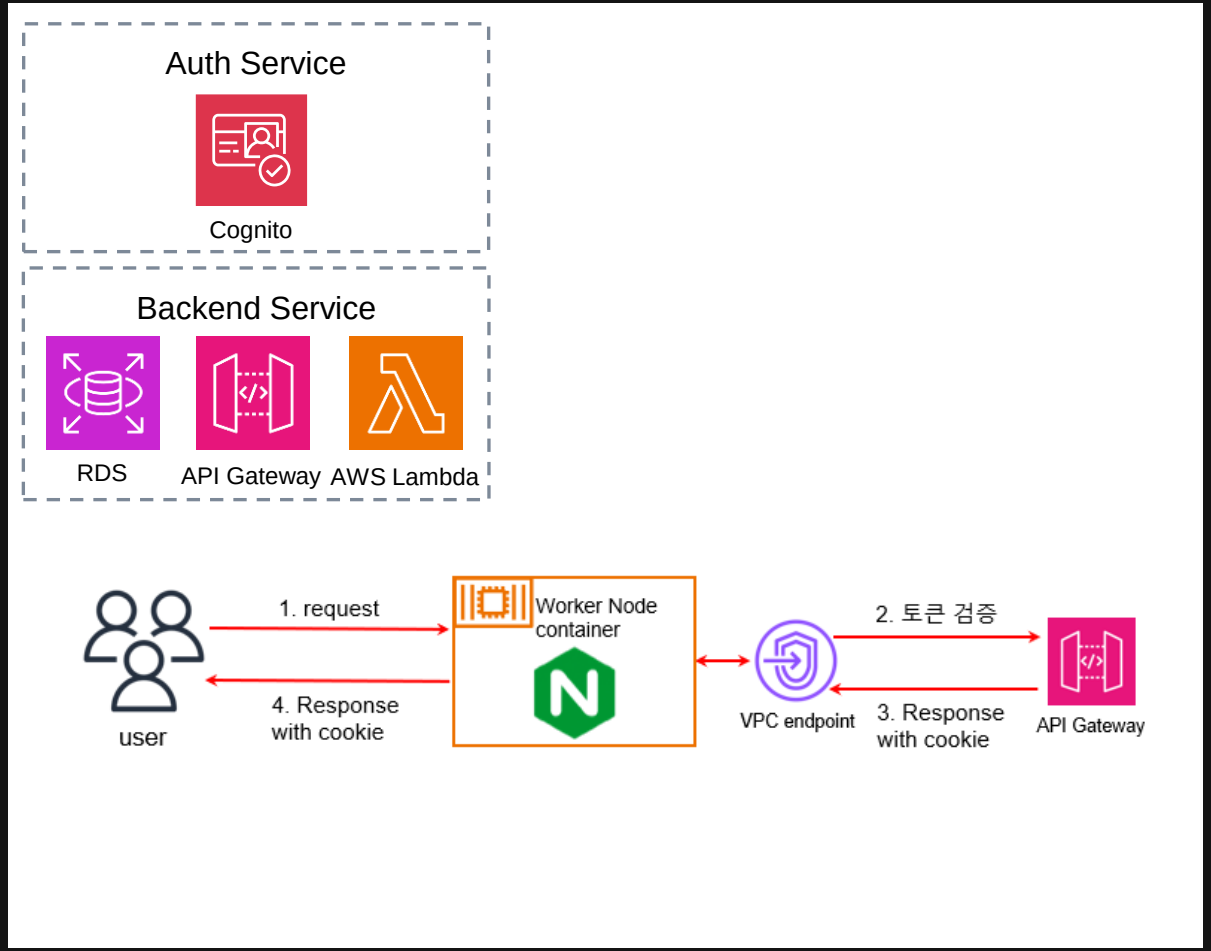

Auth & Security Service
Cognito
AWS WAF
Frontend Service
AWS EKS
Backend Service
RDS
API Gateway
AWS Lambda
Storage Service
S3
CloudFront
Cognito
user
1. request
4. Response
with cookie
Worker Node
container
VPC endpoint
2. 토큰 검증
3. Response
with cookie
API Gateway
If. 토큰 OK
Lambda
function
MySQL
RDS MySQL
instance
5. Request
with cookie
Edge location
Amazon
CloudFront
bucket

WEB SERVICE

사용자 인증



백엔드 접근 인증, Api 외부 노출 방지

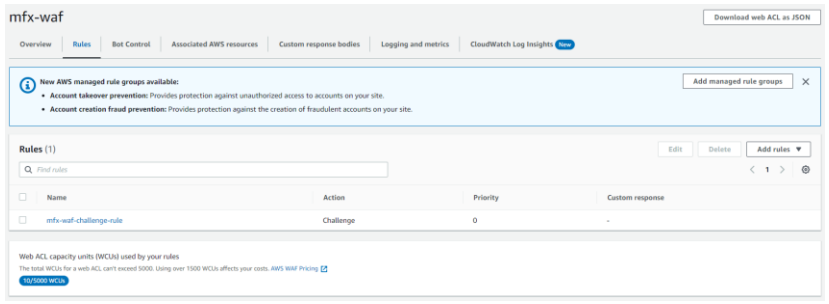


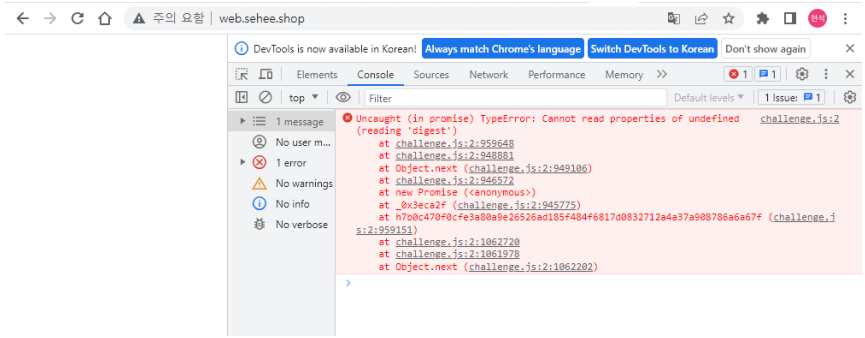
WEB SERVICE

Bot에 의한 트래픽 부하 및 각종 위협을 방지

Security Service

AWS WAF





Signed Cookie를 통한 접근 제어

Storage Service

S3

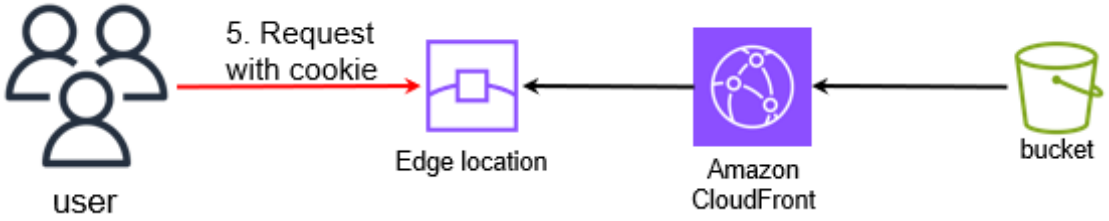
CloudFront

Backend Service

RDS

API Gateway

AWS Lambda





자동화 및 배포

- TERRAFORM
- EKS

자동화 및 배포

Amazon EKS

AWS Kubernetes 클러스터 관리 서비스



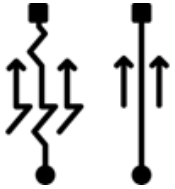
관리 용이성



보안성

Transit Gateway

다른 VPC간에 통신이 가능하게 하는 서비스



네트워크 간소화



확장성

Terraform

오픈 소스 코드형 인프라스트럭처



안정성과 신뢰성



리소스 관리의 편의성

자동화 및 배포

- Cluster
- Nodegroup
- OIDC Provider

- EKS Cluster
- Node group
- EKS Add-on

Bastion

Local

kubectl

sa

ing

svc

deploy

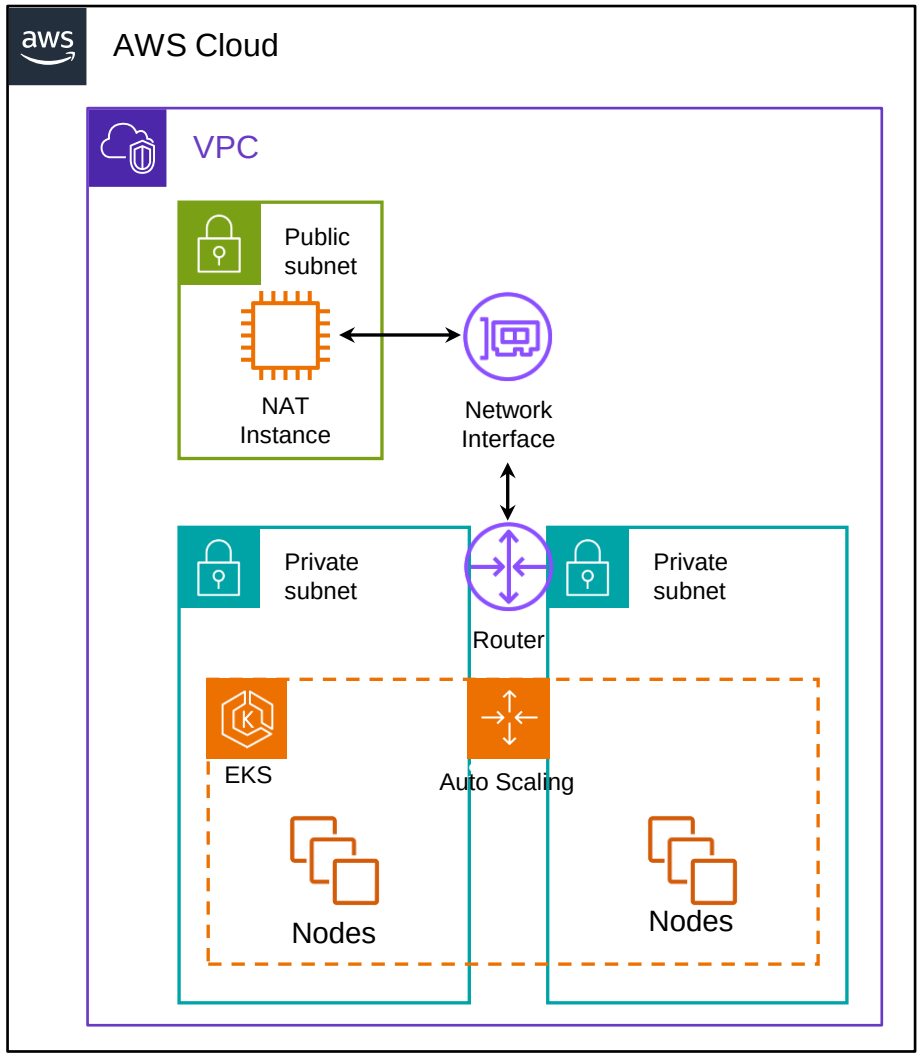
pods

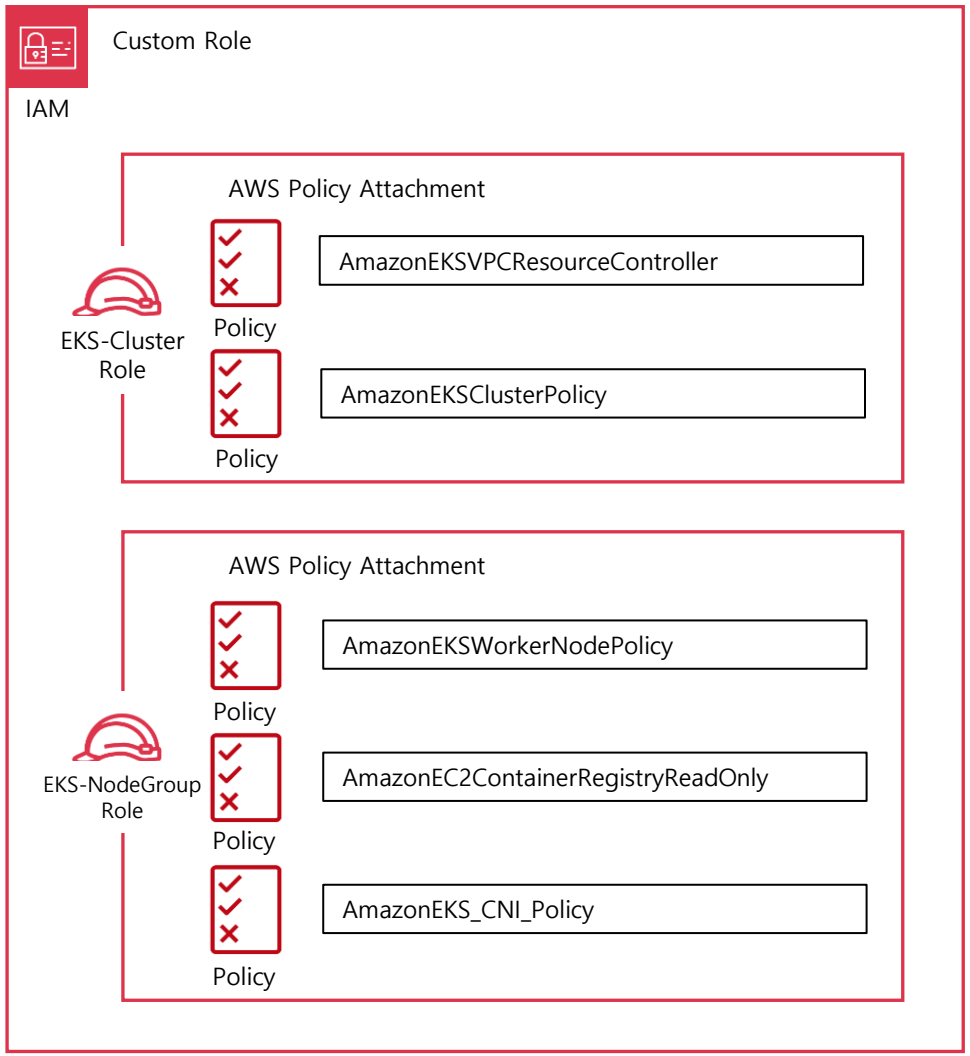
HELM

ALB Ingress Controller

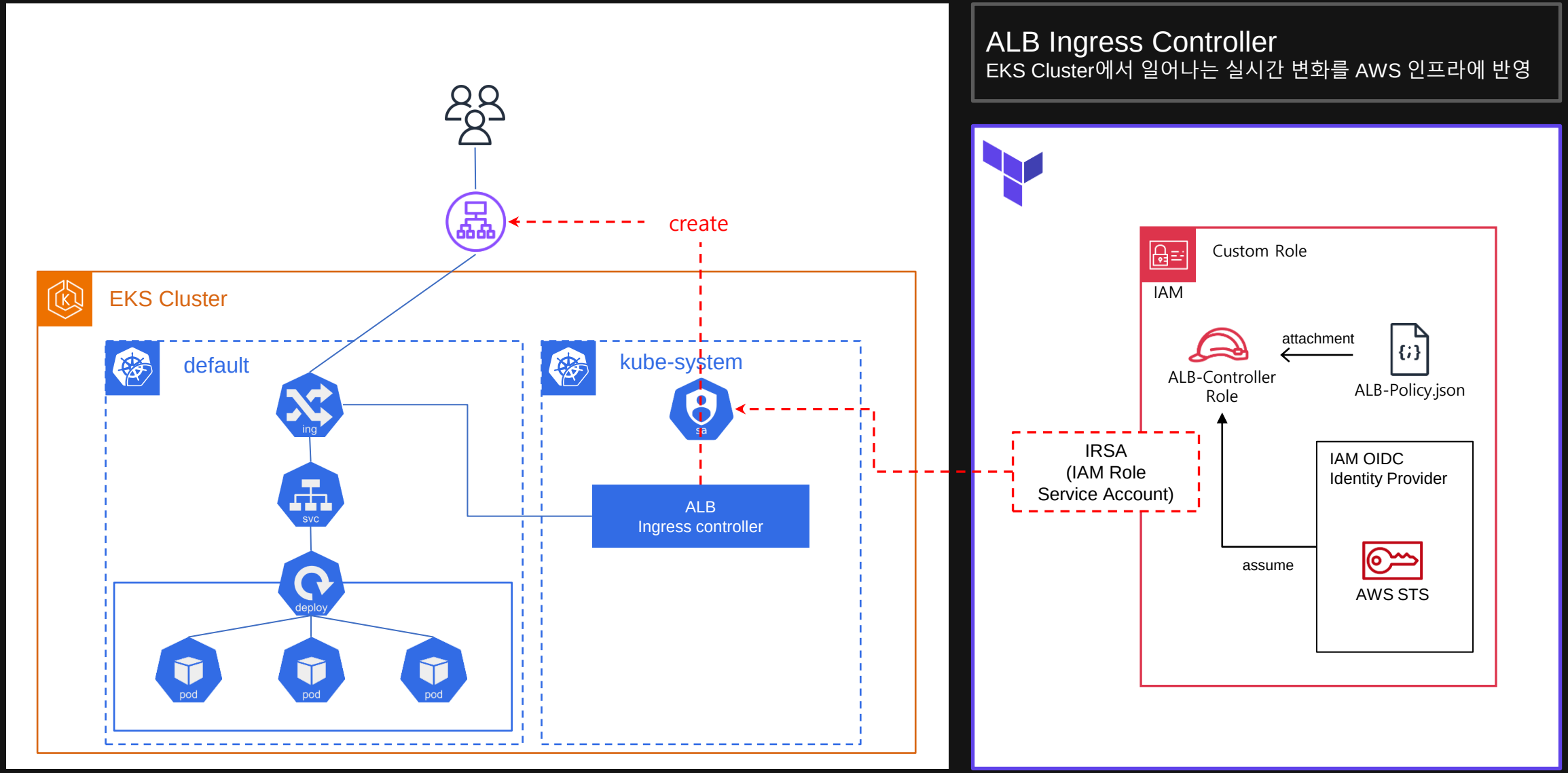
EKS AutoScaling

TERRAFORM

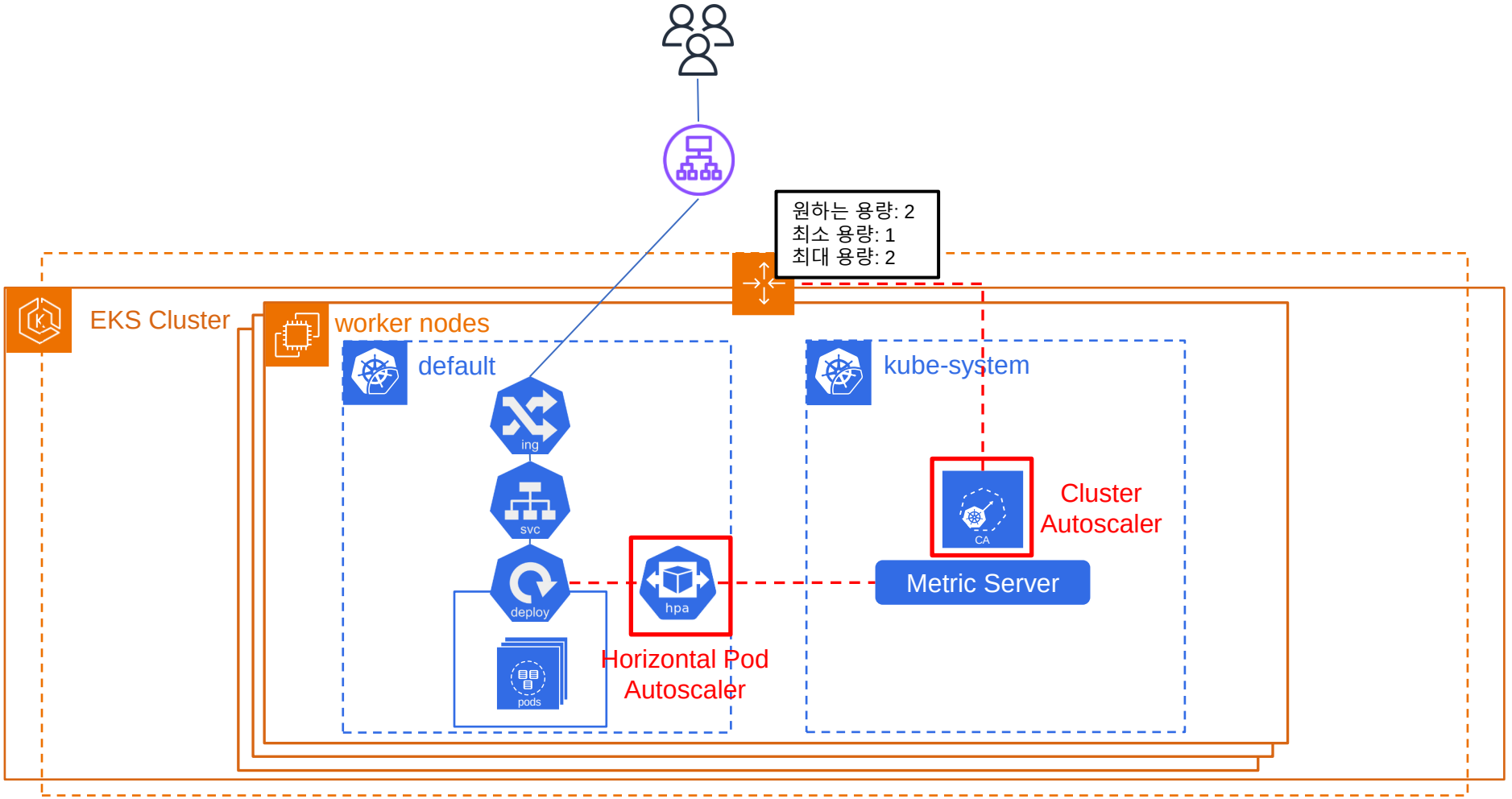

aws
AWS Cloud
VPC
Public subnet
NAT Instance
Network Interface
Private subnet
Router
Private subnet
EKS
Auto Scaling
Nodes
Nodes


Custom Role
IAM
AWS Policy Attachment
EKS-Cluster Role
Policy
AmazonEKSVPCResourceController
Policy
AmazonEKSClusterPolicy
AWS Policy Attachment
EKS-NodeGroup Role
Policy
AmazonEKSWorkerNodePolicy
Policy
AmazonEC2ContainerRegistryReadOnly
Policy
AmazonEKS_CNI_Policy

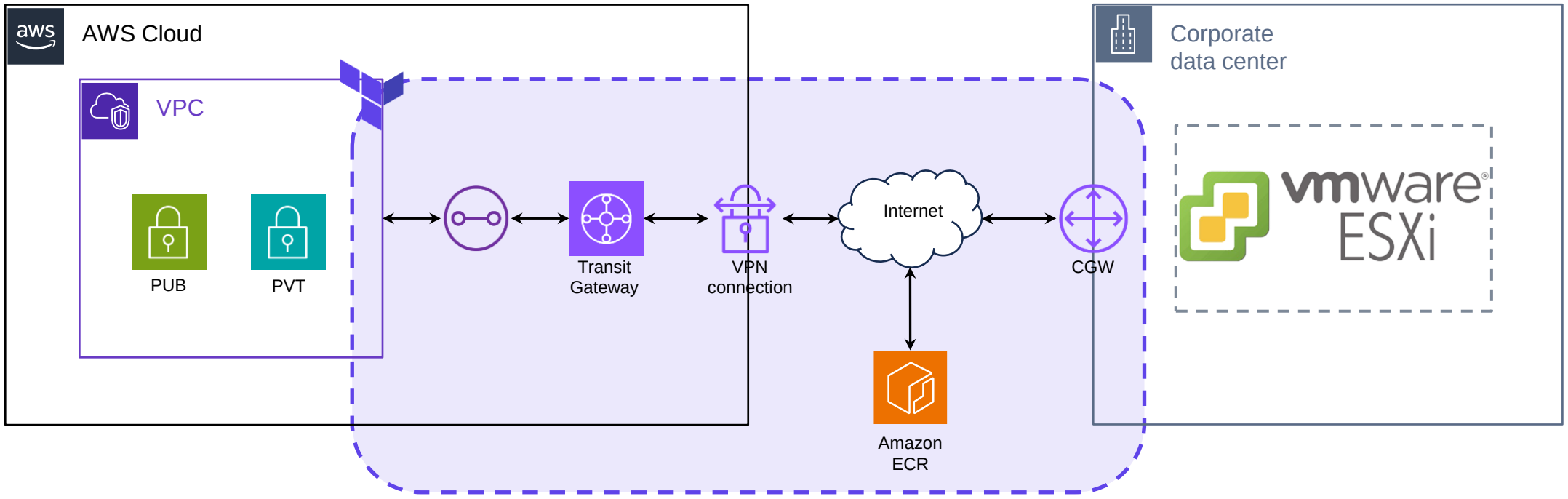
K8S - ALB INGRESS CONTROLLER



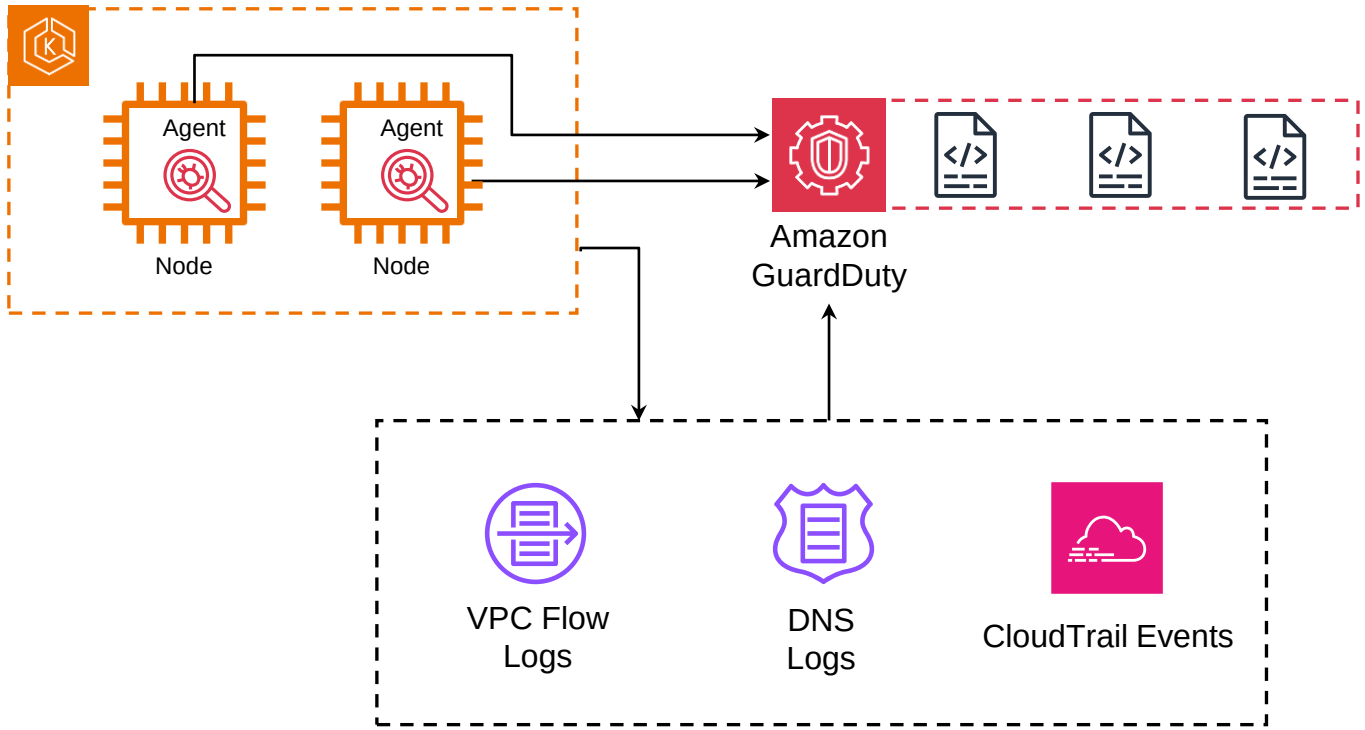
EKS - AUTOSCALING



TERRAFORM - TRANSIT GAETWAY



GUARDDUTY

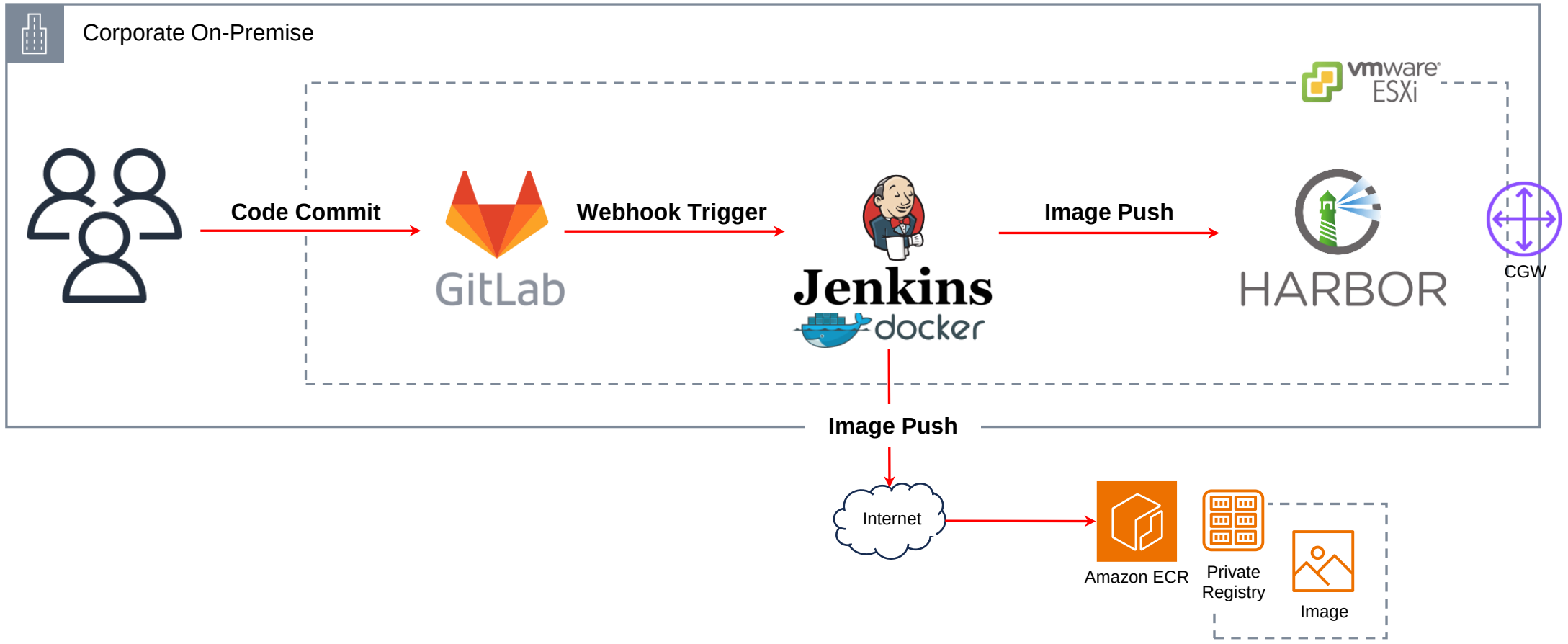

Agent
Agent
Node
Node
Amazon
GuardDuty
VPC Flow
Logs
DNS
Logs
CloudTrail Events



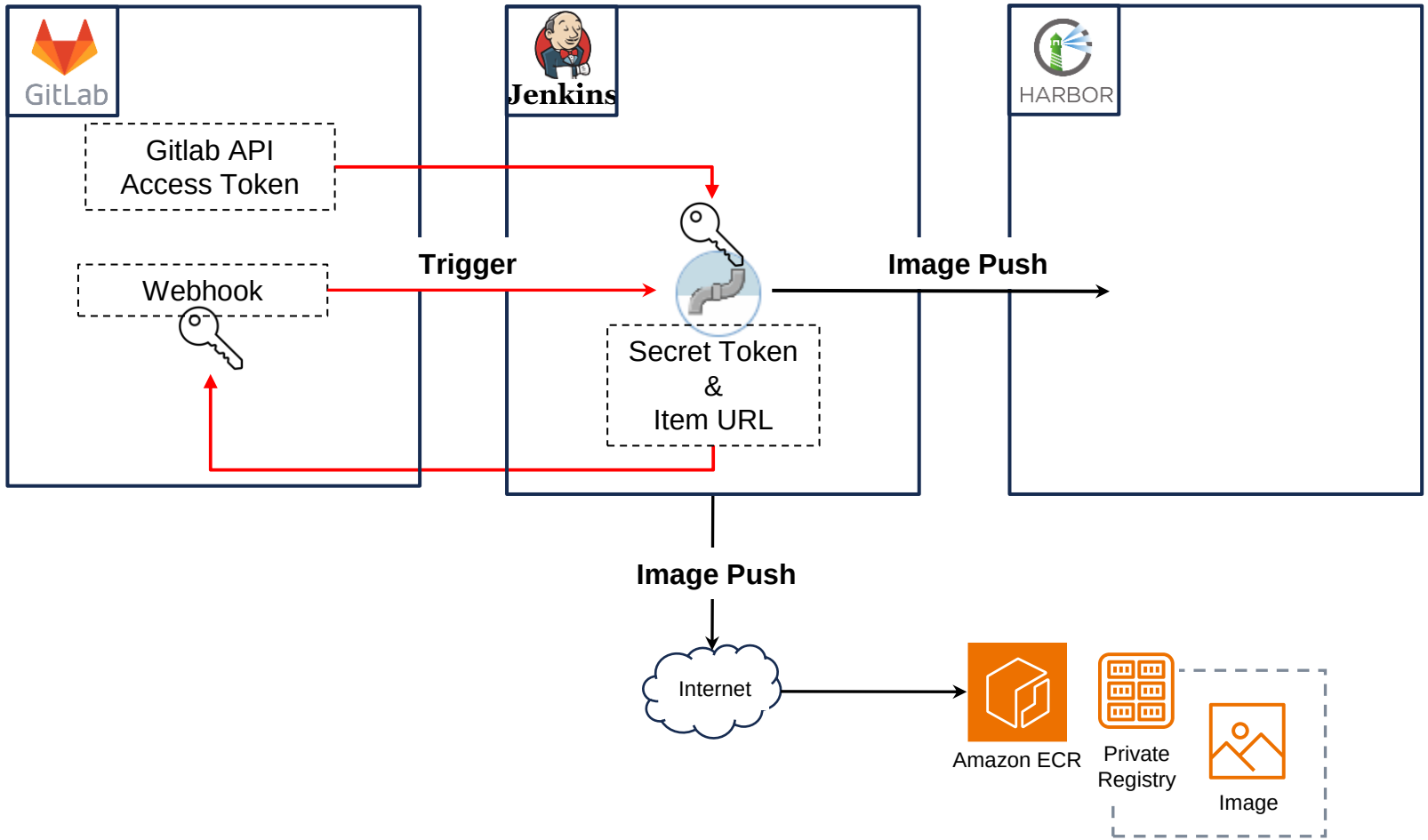
CI/CD & MONITORING

- CI/CD PIPELINE
- CLOUDWATCH

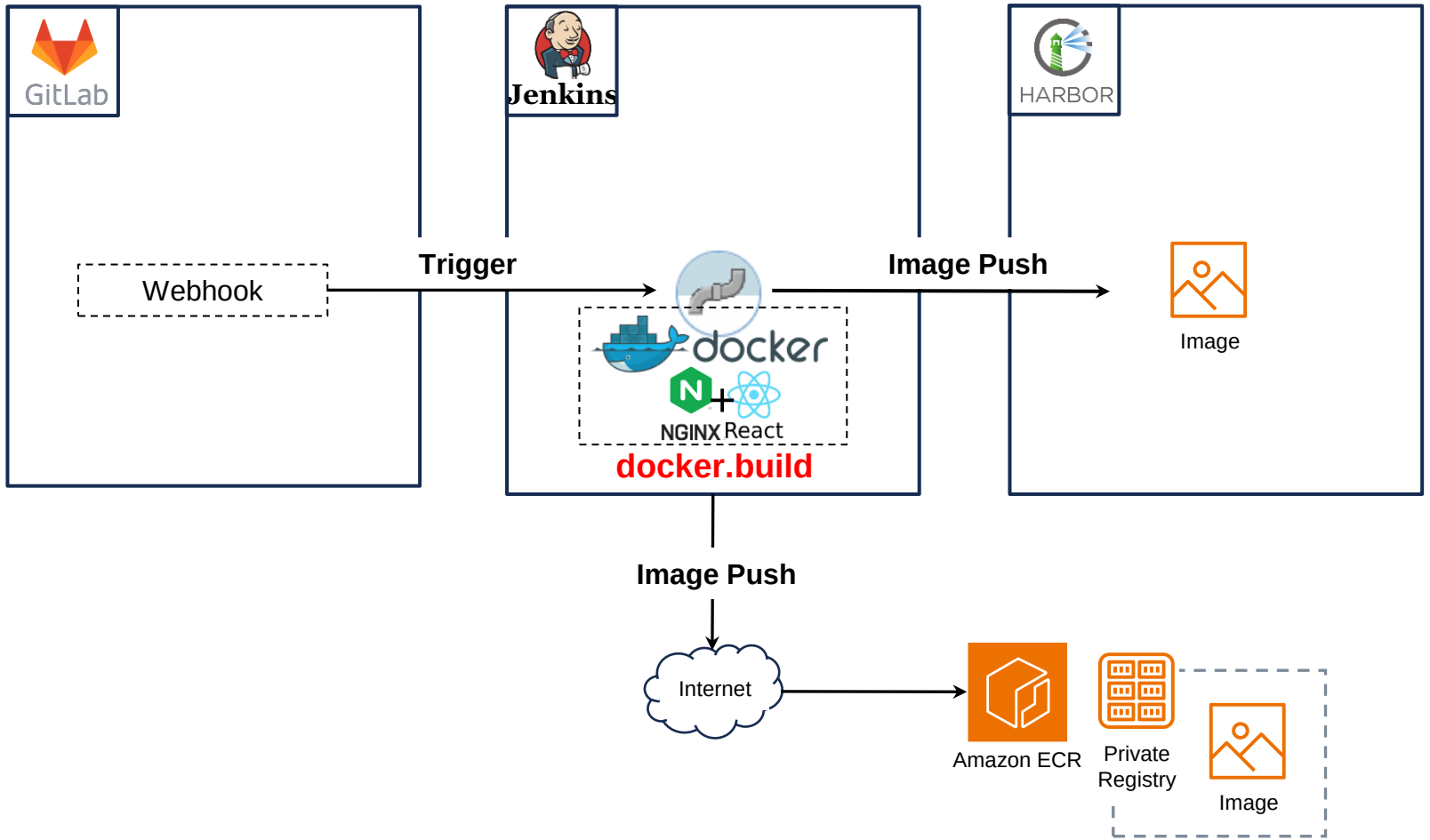
CI/CD


Corporate On-Premise
vmware ESXi
Code Commit
GitLab
Webhook Trigger
Jenkins
docker
Image Push
HARBOR
CGW
Image Push
Internet
Amazon ECR
Private Registry
Image

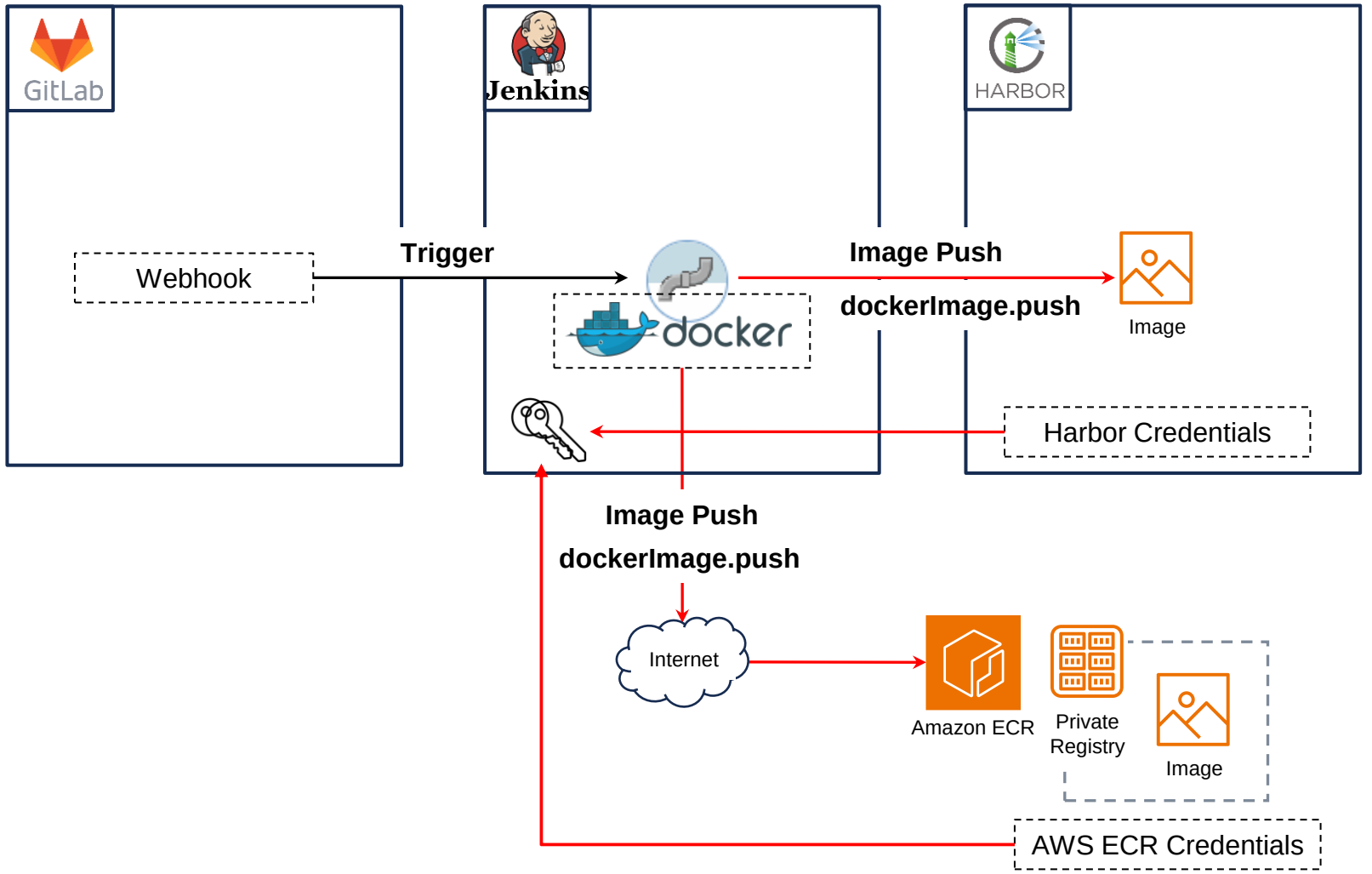
CI/CD


GitLab
Jenkins
HARBOR
Gitlab API
Access Token
Webhook
Trigger
Image Push
Secret Token
&
Item URL
Image Push
Internet
Amazon ECR
Private
Registry
Image

CI/CD


GitLab
Jenkins
HARBOR
Webhook
Trigger
Image Push
Image
docker
NGINX React
docker.build
Image Push
Internet
Amazon ECR
Private Registry
Image

CI/CD


GitLab
Jenkins
HARBOR
Webhook
Trigger
docker
Image Push
dockerImage.push
Image
Harbor Credentials
Image Push
dockerImage.push
Internet
Amazon ECR
Private
Registry
Image
AWS ECR Credentials

MONITORING

클러스터 성능 지표



Amazon EKS



Container Insights

- Clusters
- Namespace
- Nodes
- Services
- Pods

Lambda
성능 지표 및 네트워크



AWS Lambda



Lambda Insights

- 호출 및 오류
- 지속 시간
- 제한
- 메모리 사용량
- CPU 사용량
- 네트워크 사용량



AWS X-Ray

- 네트워크 추적
- 세그먼트 타임라인
- 로그

CloudFront 배포 지표



Amazon CloudFront



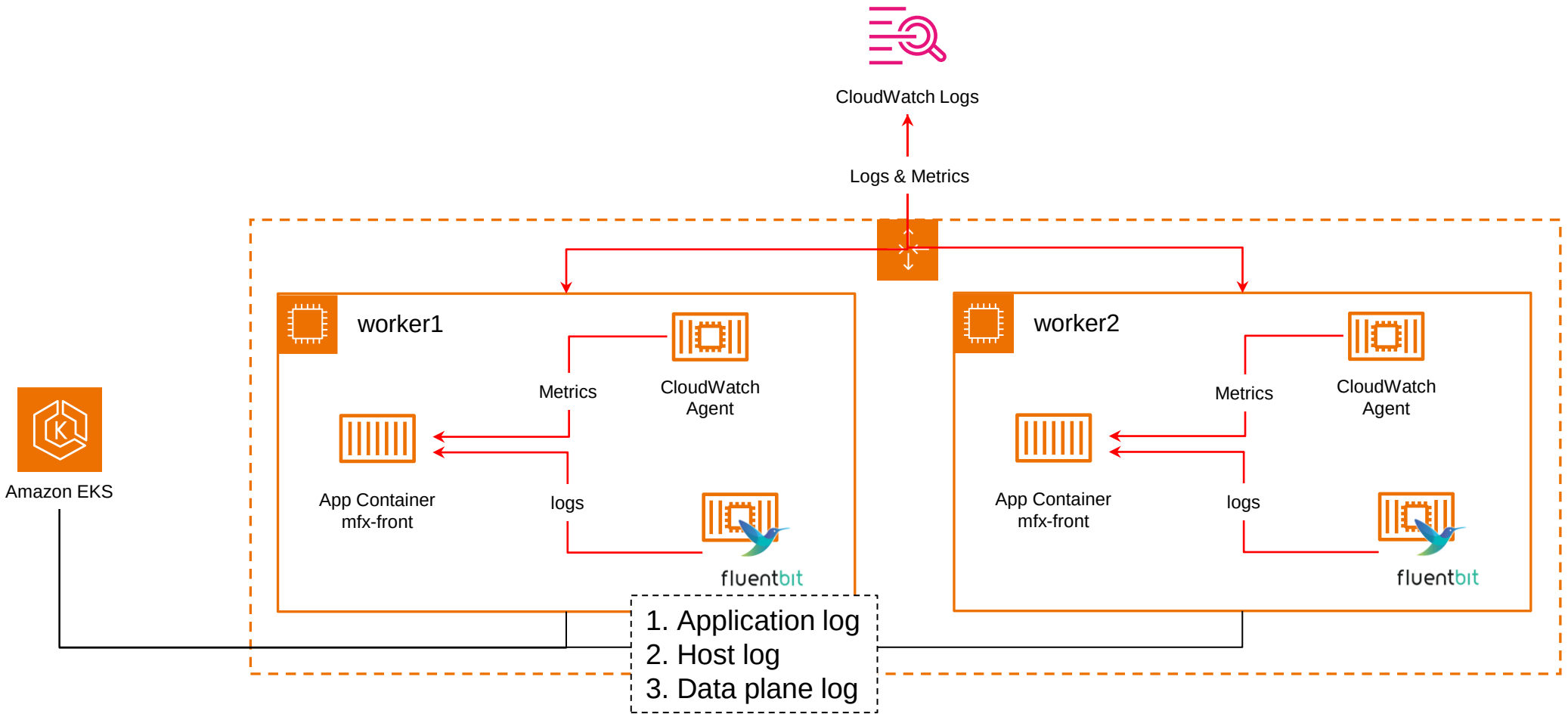
추가 배포 지표

- 요청 합
- Data transfer
- Origin latency
- Cache hit latency
- Error rate

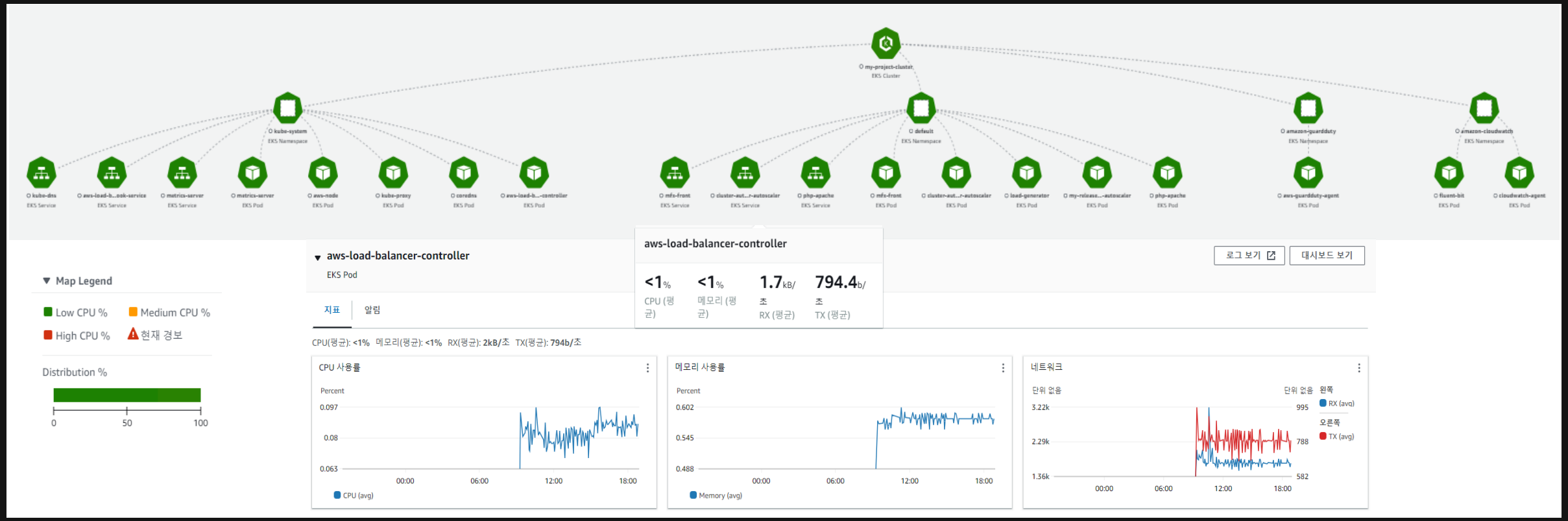


Logs

CONTAINER INSIGHTS



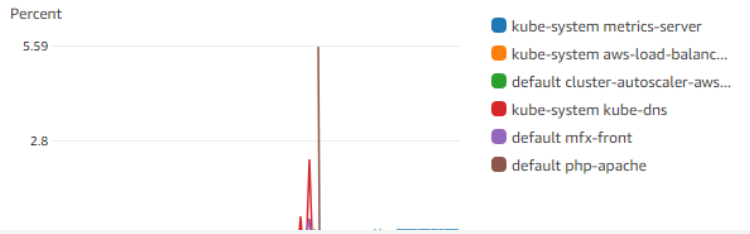
CONTAINER INSIGHTS



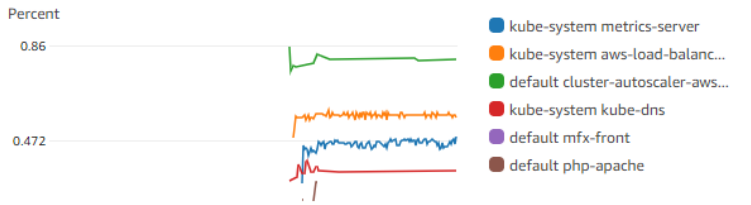
CONTAINER INSIGHTS

EKS Services

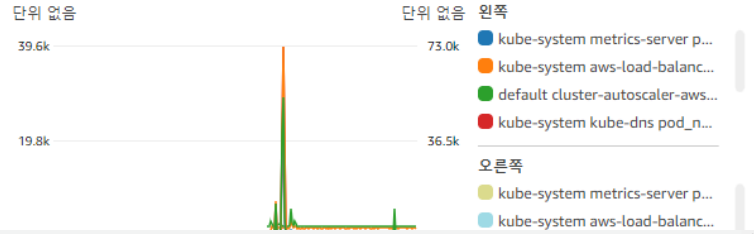
Pod CPU Utilization



Pod Memory Utilization

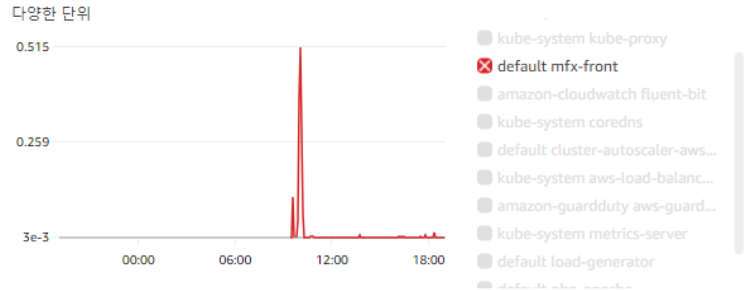


네트워크

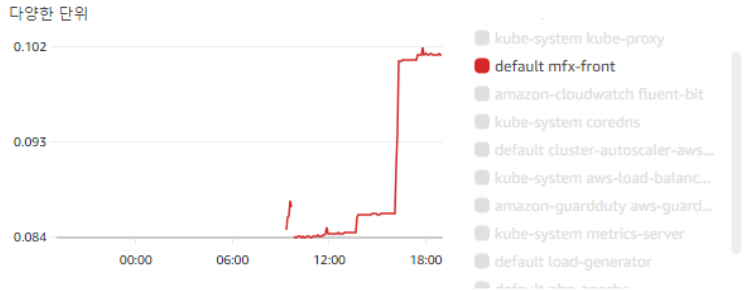


EKS Pods

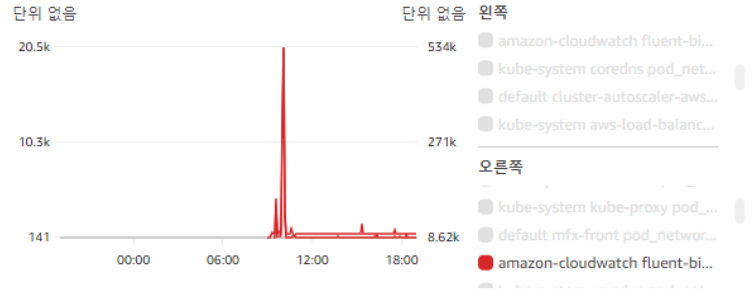
Pod CPU Utilization



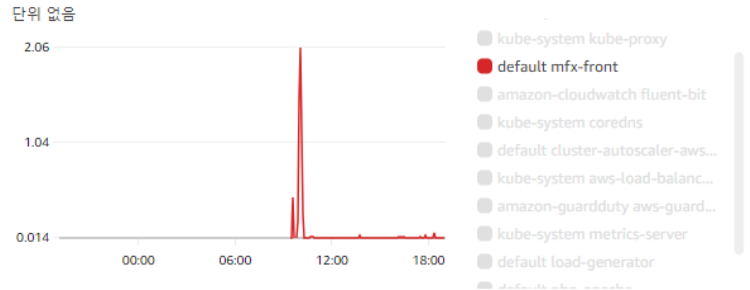
Pod Memory Utilization



네트워크



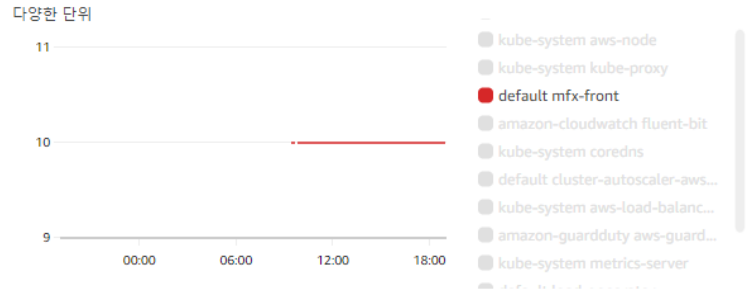
Pod CPU Utilization over Pod Limit



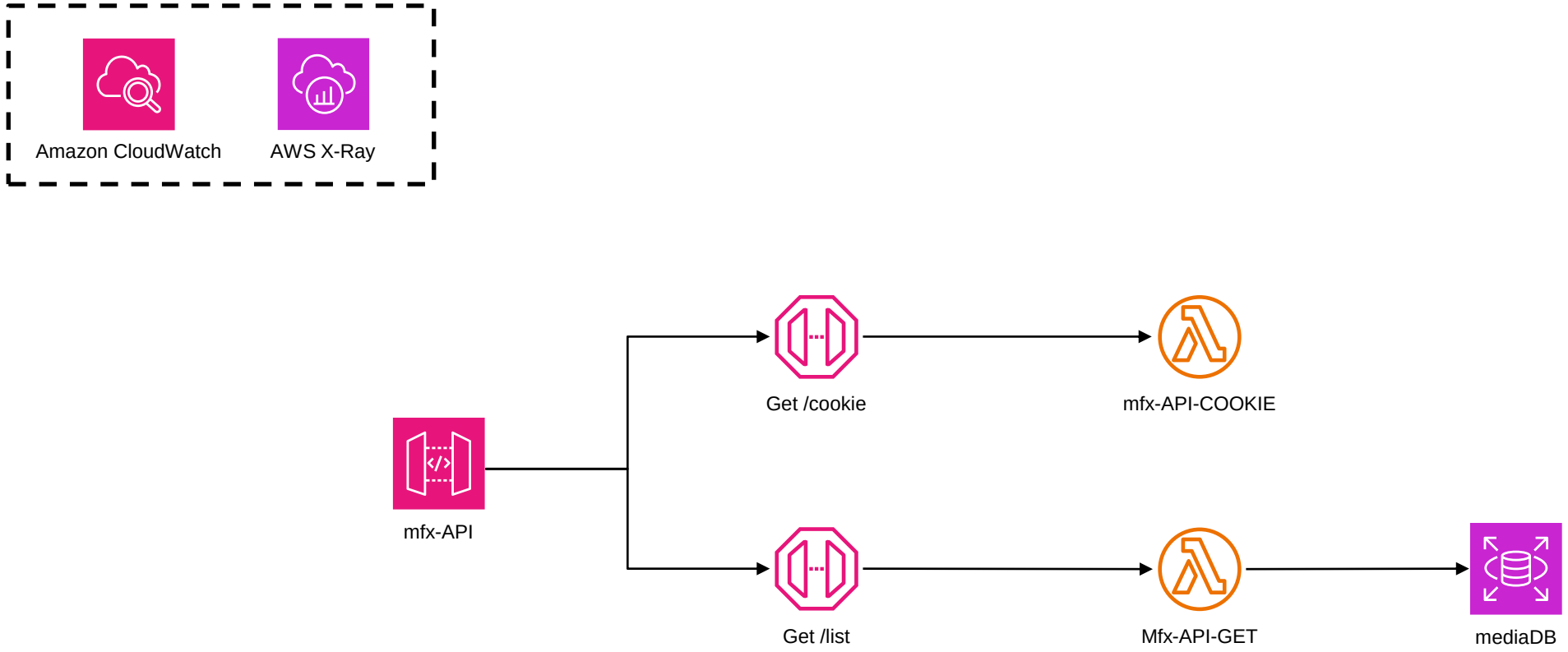
Pod Memory Utilization over Pod Limit



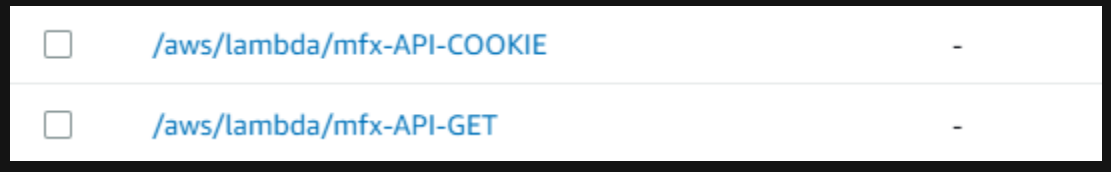
예약된 CPU 컴퓨팅 용량



LAMBDA MONITORING



LAMBDA INSIGHTS



2023-07-24T20:43:14.189+09:00 LOGS Name: cloudwatch_lambda_agent State: Subscribed Types: [Platform]

2023-07-24T20:43:14.915+09:00 EXTENSION Name: cloudwatch_lambda_agent State: Ready Events: [INVOKE, SHUTDOWN]

2023-07-24T20:43:14.955+09:00 START RequestId: 9d0a04e2-179a-4690-889a-d49cdb418a3b Version: $LATEST

2023-07-24T20:43:14.956+09:00 [2023-07-24T11:43:14.956Z INFO cloudwatch_lambda_agent::runtime_event_collector] Received next response from runtime api.

2023-07-24T20:43:15.035+09:00 [2023-07-24T11:43:15.035Z INFO cloudwatch_lambda_agent::event_sink] Received invoke event for request 9d0a04e2-179a-4690-889a-d49cdb418a3b. Map size: 0

2023-07-24T20:43:15.035+09:00 [2023-07-24T11:43:15.035Z INFO cloudwatch_lambda_agent::event_sink] Completed match. Map size: 1

2023-07-24T20:43:15.102+09:00 조회 데이터 : (('thumbnail/FastX.jpg', 'FastX', 'FastX', 'vod/hls/FastX/FastX.m3u8'), ('thumbnail/TopGun_Maverick.webp', 'TopGun_Maverick', 'TopGun_Maverick', 'vod/hls/TopGun_Maverick/…

조회 데이터 : (('thumbnail/FastX.jpg', 'FastX', 'FastX', 'vod/hls/FastX/FastX.m3u8'), ('thumbnail/TopGun_Maverick.webp', 'TopGun_Maverick', 'TopGun_Maverick', 'vod/hls/TopGun_Maverick/TopGun_Maverick.m3u8'), ('thumbnail/Elemental.webp', 'Elemental', 'Elemental', 'vod/hls/Elemental/Elemental.m3u8'), ('thumbnail/Avatar2_The_Way_of_Water.webp', 'Avatar2_The_Way_of_Water', 'Avatar2_The_Way_of_Water', 'vod/hls/Avatar2_The_Way_of_Water/Avatar2_The_Way_of_Water.m3u8'), ('thumbnail/Indiana_Jones_and_the_Dial_of_Destiny.webp', 'Indiana_Jones_and_the_Dial_of_Destiny', 'Indiana_Jones_and_the_Dial_of_Destiny', 'vod/hls/Indiana_Jones_and_the_Dial_of_Destiny/Indiana_Jones_and_the_Dial_of_Destiny.m3u8'), ('thumbnail/SpiderMan_Across_the_SpiderVerse.jpg', 'SpiderMan_Across_the_SpiderVerse', 'SpiderMan_Across_the_SpiderVerse', 'vod/hls/SpiderMan_Across_the_SpiderVerse/SpiderMan_Across_the_SpiderVerse.m3u8'), ('thumbnail/TopGun_Maverick_beach.webp', 'TopGun_Maverick_beach', 'TopGun_Maverick_beach', 'vod/hls/TopGun_Maverick_beach/TopGun_Maverick_beach.m3u8'), ('thumbnail/Guardians_of_Galaxy_Volume_3.webp', 'Guardians_of_Galaxy_Volume_3', 'Guardians_of_Galaxy_Volume_3', 'vod/hls/Guardians_of_Galaxy_Volume_3/Guardians_of_Galaxy_Volume_3.m3u8'))

복사

CLOUDFRONT MONITORING

CLOUDFRONT 추가 지표



CLOUDFRONT MONITORING

CLOUDFRONT LOGS

☐	E33ZRZ843ZG1GX.2023-07-22-06.4ab6e9e0.gz	gz	2023. 7. 22. pm 3:33:12 PM KST	491.0B	Standard
☐	E33ZRZ843ZG1GX.2023-07-22-08.5f0edfdb.gz	gz	2023. 7. 22. pm 5:08:12 PM KST	665.0B	Standard
☐	E33ZRZ843ZG1GX.2023-07-22-08.a6adb7fc.gz	gz	2023. 7. 22. pm 5:13:12 PM KST	460.0B	Standard
☐	E33ZRZ843ZG1GX.2023-07-22-09.38fdf2ff.gz	gz	2023. 7. 22. pm 6:28:12 PM KST	663.0B	Standard
☐	E33ZRZ843ZG1GX.2023-07-22-15.fcc6425e.gz	gz	2023. 7. 23. am 12:38:12 AM KST	596.0B	Standard
☐	E33ZRZ843ZG1GX.2023-07-22-17.726b6ce5.gz	gz	2023. 7. 23. am 2:18:12 AM KST	463.0B	Standard

```
#Fields: date time x-edge-location sc-bytes c-ip cs-method cs(Host) cs-uri-stem sc-status cs(Referer) cs(User-Agent) cs-uri-query cs(Cookie) x-edge-result-type x-edge-request-id x-host-header cs-protocol
2023-07-31  02:55:03    ICN54-C1    243976  106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/FastX.jpg    200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20Android%206.0;%20Nexus%205%20Bui
2023-07-31  02:55:03    ICN54-C1    76872   106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/TopGun_Maverick.webp 200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20Android%206.0;%20Nexus%2
2023-07-31  02:55:03    ICN54-C1    285066  106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/Elemental.webp   200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20Android%206.0;%20Nexus%205%2
2023-07-31  02:55:03    ICN54-C1    53285   106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/Avatar2_The_Way_of_Water.webp    200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20Android%206.
2023-07-31  02:55:03    ICN54-C1    371080  106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/Indiana_Jones_and_the_Dial_of_Destiny.webp   200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20
2023-07-31  02:55:03    ICN54-C1    264089  106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/SpiderMan_Across_the_SpiderVerse.jpg 200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20Android%
2023-07-31  02:55:03    ICN54-C1    114262  106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/TopGun_Maverick_beach.webp   200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20Android%206.0;%2
2023-07-31  02:55:03    ICN54-C1    263643  106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /thumbnail/Guardians_of_Galaxy_Volume_3.webp    200 https://web.sehee.shop/ Mozilla/5.0%20(Linux;%20Android%
2023-07-31  02:57:57    ICN54-C1    999 106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /index/index.html   200 https://web.sehee.shop/ Mozilla/5.0%20(Windows%20NT%2010.0;%20Win64;%20x64)%20AppleWebKi
2023-07-31  02:57:58    ICN54-C1    333 106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /favicon.ico    403 https://video.sehee.shop/index/index.html?link=https%253A%252F%252Fvideo.sehee.shop%252Fvod%
2023-07-31  02:57:58    ICN54-C1    758 106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /vod/hls/TopGun_Maverick/TopGun_Maverick.m3u8   200 https://video.sehee.shop/index/index.html?link=https%253A%25
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2023-07-31  02:57:58    ICN54-C1    1572605 106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /vod/hls/TopGun_Maverick/TopGun_Maverick1280x720_4mbps_qvbr_00001.ts    200 https://video.sehee.shop/index/i
2023-07-31  02:57:58    ICN54-C1    334 106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /favicon.ico    403 https://video.sehee.shop/index/index.html?link=https%253A%252F%252Fvideo.sehee.shop%252Fvod%
2023-07-31  02:57:58    ICN54-C1    1328    106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /vod/hls/TopGun_Maverick/TopGun_Maverick1920x1080_8mbps_qvbr.m3u8   200 https://video.sehee.shop/index/index
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2023-07-31  02:57:59    ICN54-C1    2743996 106.253.56.124  GET d2nprvommh7h07.cloudfront.net   /vod/hls/TopGun_Maverick/TopGun_Maverick1920x1080_8mbps_qvbr_00003.ts   200 https://video.sehee.shop/index/i
```

MONITORING

EKS 자원 경보

지표 (3,583)

모든 지표, 측정기준, 리소스 ID 또는 계정 ID 검색

▼ 사용자 지정 네임스페이스

ContainerInsights	274
LambdaInsights	18

모두 > ContainerInsights > ClusterName, InstanceId, NodeN...

모든 지표, 측정기준, 리소스 ID 또는 계정 ID 검색

ClusterName 100/104	InstanceId	NodeName	지표 이름
my-project-cluster	i-0fa271ece595e9b91	ip-10-4-64-229.ap-northeast-2.compute.internal	node_cpu_reserved_capacity
my-project-cluster	i-0fa271ece595e9b91	ip-10-4-64-229.ap-northeast-2.compute.internal	node_network_total_bytes
my-project-cluster	i-0fa271ece595e9b91	ip-10-4-64-229.ap-northeast-2.compute.internal	node_cpu_utilization
my-project-cluster	i-010f5d5118aed2d0e	ip-10-4-96-32.ap-northeast-2.compute.internal	node_cpu_reserved_capacity
my-project-cluster	i-010f5d5118aed2d0e	ip-10-4-96-32.ap-northeast-2.compute.internal	node_network_total_bytes
my-project-cluster	i-010f5d5118aed2d0e	ip-10-4-96-32.ap-northeast-2.compute.internal	node_cpu_utilization
my-project-cluster	i-010f5d5118aed2d0e	ip-10-4-96-32.ap-northeast-2.compute.internal	node_memory_utilization